

Residence Where Transits Were Design Intent: A Cross-Substrate Formalism for Coupling-Failure Pathology

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Abstract

Many biological, cognitive, and artificial systems enter transient states that are normal when briefly sampled but pathological when residence persists. We introduce a two-axis residence-time observable $R(s, c) = \tau_R(s) \times \pi_c$, combining temporal dwell τ_R in state s with compartmental occupancy probability π_c , paired with an exit-rate variable $\lambda_{\text{exit}}(s, c) = 1/\tau_R(s, c)$ that captures the return-path-machinery capacity. Pathology corresponds to R escaping substrate-specific functional residence bounds on either axis, with s and c possibly unchanged; a normalized form $\Delta_R(s, c) = \log(\tau_R/\tau_0) + \log(\pi_c/\pi_0)$ makes the observable substrate-portable. Four canonical disease cases (FUS in stress granules, PrP topology, Huntingtin nuclear-versus-mitochondrial residence, tau multi-compartment mislocalization) establish that two axes are empirically necessary: single-axis observables collapse clinically relevant distinctions. We demonstrate the formalism across eight substrates spanning receptor pharmacology, proteostasis, cancer phospho-regulation, autonomic physiology, exercise physiology, large-scale brain network dynamics, memory phenomenology and public-epistemic information ecosystems, and recursive AI/data training loops. We position the contribution as the formal-operational composition of relation-first prior art (Rovelli, French & Ladyman, Maturana & Varela, Friston, Carhart-Harris-Chandaria-Friston, Whitehead, Simondon, Copeland) into one empirically tractable formalism with a substrate-uniform return-path-failure pathology grammar, not as a first articulation of relation-first ontology at any single substrate. A cross-substrate experimental program of thirteen experiments with binary falsification conditions, organized into recovery-kinetics and pharmacology-to-psychedelics-bridge clusters, is proposed; five are designated for OSF pre-registration.

1. Introduction

The claim that relations are ontologically prior to relata — that coupling or process is the substrate and components are derived from it — has been made, in various forms, by philosophers and scientists working in at least five distinct research traditions over the past century. In fundamental physics, Rovelli's relational quantum mechanics treats quantum states as existing only relative to other systems, with no absolute state (Rovelli 1996; Di Biagio & Rovelli 2022; Calosi & Riedel 2024). Ontic structural realism argues that fundamental ontology is structure rather than objects with properties, with objects being nodes within relational structure (French & Ladyman 2003; Ladyman & Ross 2007; French 2014). Karen Barad's agential realism denies that there are pre-existing independent relata and treats phenomena as primary (Barad 2007), with extensions into psychology (Scholz 2024) and social theory (Zanotti 2025). In biology, Maturana and Varela's autopoiesis treats life as self-producing organization

rather than as a property of components (Maturana & Varela 1980). Rosen's relational biology formalizes organisms as categories of relations with closure-to-efficient-causation (Rosen 1991). Mossio and Montévil's closure-of-constraints framework makes the formal biology-of-relation move most rigorously (Montévil & Mossio 2015; Bich 2024). And Friston's free-energy principle treats particles, organisms, agents, and communicating ensembles within a common Markov-blanket framework (Friston 2010; Friston, Levin, Sengupta & Pezzulo 2015; Kuchling et al. 2020; Ramstead, Badcock & Friston 2018; Friston et al. 2024; Friston et al. 2025). At the brain-substrate level, canalization theory unifies depression, obsessive-compulsive disorder, addiction, post-traumatic stress disorder, eating disorders, psychosis, and somatoform conditions under a single principle of cognitive-and-behavioral phenotypes as canalized features of mind, brain, or behavior that have come to dominate an individual's psychological state space (Carhart-Harris, Chandaria, Friston et al. 2023), with refinements in Deep CANALs work (Juliani, Safron & Kanai 2024). And philosophically, Whitehead (1929), Simondon (1958), and the dependent-origination tradition in Buddhist philosophy have articulated process-first and relation-first ontologies in various forms, with recent formal engagement extending these positions (Stenner 2024; Del Fabbro & Weaver 2025; Ho et al. 2022).

These are distinct research programs operating on different substrates with different methods. What they share is a commitment to treating relation, process, or coupling as ontologically prior to whatever might be taken as the discrete components of the system in question.

What has *not* been established across these traditions is a single explicit measurable coupling-first observable. Each substrate-level program either stays at the conceptual or metaphysical level (Whitehead, Simondon, Barad) or provides formal machinery that applies within its domain without extending as a single observable across substrates (Rovelli within quantum; Maturana-Varela within biology of cognition; Mossio within organizational biology; Carhart-Harris within psychopathology). Friston's free-energy principle comes closest to a substrate-general formalism but in his published work stops short of the explicit ontological claim that coupling is primary across all substrates uniformly — his characteristic idiom is that systems “can be described as” or “look as if” they minimize free energy, which is methodological rather than ontological commitment.

Nor has any of these programs produced what we will call a *downstream pathology grammar*: a principled account of what goes wrong when coupling structure fails, stated at the formal level and applied uniformly across substrates. Canalization theory comes closest at the brain substrate, and it has profoundly influenced this paper's approach, but it is not applied as a cross-substrate principle outside of psychopathology.

The contribution of this paper is the formal-operational composition of these substrate-level programs into one empirically tractable formalism. We introduce a two-axis residence-time observable that operates uniformly across substrates and expresses a substrate-uniform residence-pathology principle: **pathology in any system with both regimes available to it is the temporal-and- spatial residence pattern of states whose transient sampling was the design intent, produced by failure of the return-path machinery that would have enforced return to functional control, not the identity of the states themselves.**

We are explicit about what this contribution is not. We do not claim first articulation of relation-first ontology at any substrate level; that territory is densely occupied by the authors

cited above. We do not claim first articulation of cross-disorder residence-pathology at the brain substrate; canalization theory established that in 2023. We do not claim first articulation of the three-regime structure of human–AI coupled dynamics, which Wu, Kang, Xu, Xie, Mi, Wang, Liu & Chen (2026, arXiv:2605.06347) have independently proposed in minimal-ODE form during the preparation of this paper, and which we cite as proximate prior art for the AI/data-substrate instance of the framework’s principle. We do not claim the G_2 / octonion mathematical-physics interpretation of the observable, which we defer to a companion paper (Graise, in preparation) because the required derivation work belongs in a separate treatment. What we do claim is the explicit integration: the formal observable (with its exit-rate and normalized cross-substrate forms), the substrate-uniform symptom grammar, and the cross-substrate demonstration that a single formalism can describe coupling-failure pathology across domains as different as receptor binding, protein aggregation, brain-network dynamics, memory phenomenology, public-epistemic attractor formation, and recursive generative-model training.

This claim is narrower than some formulations of the framework’s thesis might suggest. It is also stronger for being narrower. Six adversarial prior-art surveys conducted in preparation for this paper (documented in the framework’s technical archive) have repeatedly forced the concession that substrate-level relation- first ontology is not the contribution; what remains genuinely open after those surveys is the formal-operational composition with the individual-level observable and the pathology grammar. That is what this paper delivers.

The paper is organized as follows. Section 2 defines the two-axis observable $R(s,c) = \tau_R(s) \times \pi_c$ and establishes the machinery needed to measure it. Section 3 presents four canonical disease cases — FUS in stress granules, PrP topology, Huntingtin nuclear- versus-mitochondrial residence, and tau multi-compartment mislocalization — that demonstrate both axes of the observable are necessary rather than optional. Section 4 states the cross-class unifying claim. Section 5 presents the symmetric formulation of kinetic stabilization. Section 6 identifies return-path machinery as a multiscale class across four substrates. Section 7 extends Wolynes-Ferreiro frustration theory to residence-shaped rather than state-shaped pathology. Section 8 demonstrates the cross- substrate extensions in seven further domains. Section 9 differentiates the framework from adjacent literature. Section 10 proposes an audit-design template for cross-substrate claims. Section 11 proposes thirteen experiments as a cross-domain empirical program. Section 12 discusses implications. Section 13 states non-goals. Section 14 acknowledges the deferred companion paper.

2. The Two-Axis Residence-Time Observable

2.1 Definition

For a system whose dynamics can be described on a Markov state model over conformational or functional states $s \in \mathcal{S}$, and whose components can occupy compartments $c \in \mathcal{C}$, we define the **two-axis residence observable**:

$$[R(s, c) = \tau_R(s) \pi_c]$$

where:

- $(\tau_R(s))$ is the **temporal residence time** in state s , measured as the mean dwell time on the Markov state model of native dynamics. Single-subject estimation machinery exists in the neuroscience literature (Vidaurre, Smith & Woolrich 2017 HMM Fisher kernel; Löhr et al. 2021 VAMPNet/Koopman MSM of A β 42; Tancredi et al. 2024 HMM dwell-time distributions for Hsp90).
- (π_c) is the **spatial residence**, or compartmental occupancy probability, measured by single-cell trafficking assays, live-cell imaging, compartment-resolved proteomics, or equivalent machinery depending on substrate. The compartmental- occupancy framework is well-developed in subcellular biology (Bertolotti 2018; Kumar & Lapierre 2021; Giandomenico et al. 2022) and in network neuroscience through regional-occupancy metrics (Kaiser et al. 2016; Vidaurre et al. 2017).

Functional residence bounds for both τ_R and π_c are substrate-specific ranges established by evolved or engineered return-path machinery. A state has a functional τ_R if the system's normal operation requires dwell in s within some range $[\tau_{\min}, \tau_{\max}]$; a compartment has a functional π_c if the system's normal operation requires occupancy probability within $[\pi_{\min}, \pi_{\max}]$. These bounds are not universal constants; they are determined by the specific return-path machinery operating in the substrate of interest. We use “functional residence bounds” as the formal term and “design intent” informally where the engineered/evolved-purpose framing is clarifying; the two phrasings refer to the same quantity.

Pathology corresponds to $R(s, c)$ escaping its functional residence bounds in *either* axis, with s and c possibly unchanged. This is the critical feature: pathology is not a wrong state or a wrong compartment, but a deviation in residence pattern for states and compartments that are entirely normal in other contexts.

2.1.1 Return-path machinery and the exit-rate variable

The two-axis residence observable measures what *accumulates*. The framework's central claim is that pathological residence accumulates because the *return-path machinery* that should restore the system to functional residence has failed. The accumulation variable $R(s, c)$ and the restoration-capacity variable are formally distinct and should be reported as a coupled pair.

For each state s and compartment c we define the **exit rate**:

$$[\lambda_{\text{exit}}](s, c) = \frac{1}{\tau_R(s, c)}$$

or equivalently the **return rate constant** $k_{\text{return}}(s, c) = 1/\tau_R(s, c)$. At the molecular substrate, λ_{exit} corresponds directly to the off-rate k_{off} familiar from the Copeland residence-time pharmacology (Tummino & Copeland 2008): drugs with low k_{off} have long target-residence and high $R(s, c)$ on the target-bound state; the slow off-rate is the residence-pathology when target-bound state is pathological, or the residence-protection when target-bound state is therapeutic. At the brain-network substrate, λ_{exit} corresponds to the inverse mean dwell time on a metastable state (Vidaurre, Smith & Woolrich 2017); slow exit from a depression-associated metastable state is the network-level residence-pathology. At the autonomic substrate, λ_{exit} corresponds to the inverse recovery time constant after stressor

withdrawal (engineering-resilience formalism, van Nes & Scheffer 2007); slow vagal return after cold-pressor is the autonomic-substrate residence-pathology.

The formal residence-pathology principle, stated symmetrically:

- **Residence burden** — the quantity that accumulates: $R(s, c) = \tau_R(s) \times \pi_c$
- **Return-path capacity** — the quantity that restores: $\lambda_{\text{exit}}(s, c) = 1/\tau_R(s, c)$ on the residence-axis; analogous return-rate constants on the compartmental axis (rate of probability flux out of c)
- **Pathology condition** — burden exceeds functional bounds because return capacity has failed: $R(s, c) > R_{\text{max}}(s, c)$ where R_{max} is the substrate-specific functional ceiling implied by intact return-path machinery

This decomposition is what §6 (return-path-machinery typology) elaborates substrate-by-substrate. The four return-path machinery classes identified in §6 — proteostatic, phospho-regulatory, autonomic, network-dynamical — are the substrate-specific instances of the λ_{exit} -restoration function. Disease at any of these substrates can be read as λ_{exit} falling below the functional threshold needed to maintain $R(s, c)$ within bounds.

2.1.2 Normalized two-axis observable for cross-substrate comparison

Because τ_R has units of time (seconds for receptor binding, milliseconds for stress-granule dynamics, seconds for brain-network metastable states, minutes for autonomic recovery, days for cancer phospho-regulatory resetting, years for cognitive-aging trajectories) and π_c is dimensionless, $R(s, c)$ is substrate-specific in scale and cannot be compared directly across substrates without normalization to the substrate's own functional baseline.

We define the **normalized two-axis residence**:

$$[(s, c) =]$$

where $\tau_0(s)$ and $\pi_0(c)$ are the substrate-specific functional-baseline values — the dwell time and compartmental occupancy that an intact return-path machinery would maintain. Both axes of $()$ are dimensionless; $() = 1$ corresponds to baseline operation; $() > 1$ corresponds to residence beyond the functional ceiling on the relevant axis; $() < 1$ corresponds to residence below the functional floor.

The **log-residence deviation**:

$$[{}_R(s, c) = +]$$

decomposes the residence pathology into separable temporal and spatial contributions on a symmetric scale, with sign indicating direction (positive = over-residence; negative = under-residence). The pathology condition becomes:

$$[|{}_R(s, c)| >]$$

where ε is a substrate-specific threshold determined empirically from the dynamic range over which the return-path machinery maintains residence within functional bounds. The threshold ε is the substrate's *resilience budget*: small ε indicates a substrate where small deviations are tolerated and large ε indicates a substrate that tolerates wide swings before pathology.

This normalization makes $R(s, c)$ substrate-portable. Recovery time constants, dwell-time persistence indices, compartmental segregation indices, and other substrate-specific empirical quantities all map onto components of $(\cdot)$ and $(\cdot)_R$. The cross-substrate predictions of §11 (Experiments 1, 4, 6) are formally stated as predictions about correlations across substrates in $(\cdot)_R$ and λ_{exit} measured per subject. The audit-design template (§10) requires reporting baseline τ_0 and π_0 explicitly so that $(\cdot)$ is computable from published data.

2.2 Why two axes are necessary

The two-axis structure is empirically necessary rather than theoretically convenient. Section 3 presents four canonical disease cases. Two of them (FUS in stress granules, PrP topology) require only one axis — the temporal in the FUS case, the spatial in the PrP case. The other two (Huntingtin nuclear-versus-mitochondrial residence, tau multi-compartment mislocalization) cannot be described with a single axis: the same mutant protein produces qualitatively different pathology depending on both its conformational dwell time and its compartmental occupancy. A single-axis observable necessarily collapses these distinctions, which matters for diagnostic and therapeutic stratification.

2.3 Relation to existing formalisms

The formal machinery required to measure $R(s, c)$ already exists in the neuroscience, cell biology, and pharmacology literatures, but has not been integrated across substrates as a single observable. In single-subject neuroscience, Vidaurre et al. (2017) developed HMM Fisher kernels for individual-level dwell-time prediction of cognitive traits; Singleton et al. (2022) used receptor-informed network control theory to estimate state-transition energies at the group level; Vohryzek et al. (2024) fit whole-brain models to responder and non-responder groups under psilocybin therapy. In cell biology, Hsp90 client dwell-time distributions (Tancredi et al. 2024) and MSM-based A β 42 conformational dynamics (Löhr et al. 2021) provide τ_R at the molecular level, with compartmental occupancy π_c estimated by conventional imaging and proteomics approaches. In pharmacology, residence-time $\tau = 1/k_{\text{off}}$ is standard practice following the Copeland framework (Tummino & Copeland 2008; Bosma et al. 2017; Sykes et al. 2017).

What this paper adds is the unified two-axis formulation applied uniformly across substrates as a single empirically tractable formalism. The individual-level (single-subject, single-cell, or single-system) nature of the observable is load-bearing: group-level relational metrics exist across substrates, but none combines both axes into a single predictor of coupling-failure outcome at the individual level.

2.4 Note on mathematical structural realization

The framework's underlying mathematical structure — the symmetry of coherent coupling structures and the specific role of non-associative algebra — involves considerations that substantially exceed the scope of this paper. Existing literature uses octonion and G_2 structures in symmetry analysis, compactification, and particle-physics foundations (Baez 2002; Joyce 2000; Dubois-Violette 2016; Todorov & Dubois-Violette 2018; C. Furey 2018; Marrani, Corradetti & Zucconi 2025; Singh 2026). We have not found prior publications making the stronger ontological interpretation that we propose, and the derivation work required to establish that

interpretation rigorously belongs in a separate treatment. We defer that interpretation to a companion paper (Graise, in preparation) whose burden is to (i) derive why non-associativity should be read as irreducible relationality rather than merely algebraic failure of reassociation; (ii) show why G_2 specifically, rather than some broader exceptional or categorical structure, is the right symmetry notion for coherent coupling; and (iii) define at least one concrete invariant or observable from associators or G_2 -equivariant quantities that distinguishes coherent from failed coupling. Paper 14’s observable $R(s, c)$ does not depend on the mathematical interpretation proposed there, and is stated here in substrate-neutral terms.

3. Four Canonical Disease Cases

We present four canonical disease cases from the proteostasis literature. Each case is independently well-characterized in its own field; what the framework contributes is the observation that the two-axis observable $R(s, c) = \tau_R(s) \times \pi_c$ is *necessary* rather than optional to describe them. Two of the four cases are “pure” in the sense that one axis is approximately constant and the other axis carries the pathological signal. The other two cannot be described with a single axis at all: the same protein in the same organism produces qualitatively different diseases depending on both its conformational dwell and its compartmental occupancy. Demonstrating the joint necessity of the two axes is the empirical core of this paper.

The four cases are summarized in Table 1 before the detailed treatment of each, so that the structural argument is visible at a glance. Figure 1 plots the same four cases in the two-axis observable plane, showing how each case escapes the functional residence bounds along its own characteristic axis.

Table 1. Four canonical disease cases mapped to the two-axis residence observable.

Case	Protein / system	Temporal axis (τ_R)	Spatial axis (π_c)	Why this case requires the noted axes
3.1	FUS in stress granules	Necessary	Less central	Persistent residence in granule state (granule “aging” to gel) is the proximal pathology; granule compartmental address is approximately constant
3.2	PrP topology	Less central	Necessary	Pathological signal is membrane-topology / compartmental

Case	Protein / system	Temporal axis (τ_R)	Spatial axis (π_c)	Why this case requires the noted axes
3.3	Huntingtin nuclear vs mitochondrial residence	Necessary	Necessary	rerouting ($\pm$ TM versus secreted versus GPI- anchored); state-level conformation alone is insufficient Same mHtt protein produces qualitatively different pathology depending jointly on conformational dwell and compartmental occupancy; single-axis description collapses the distinction
3.4	Tau multi- compartment mislocalization	Necessary	Necessary	Three distinct spatial regimes (axonal, somatodendritic, extracellular) each combined with RNP- condensate residence on the temporal axis; pathology grammar requires both axes simultaneously

The pure-axis cases (3.1, 3.2) establish that each axis carries independent empirical content: a one-axis observable would either fail to detect FUS pathology (if it tracked only π_c) or PrP pathology (if it tracked only τ_R). The both-axes cases (3.3, 3.4) establish that no one-axis

observable suffices for the full clinical heterogeneity: the same protein produces distinct diseases under different combinations of dwell and occupancy that no single dimension can distinguish. The two-axis structure is therefore necessary rather than convenient. The remainder of §3 develops each case in detail.

3.1 FUS in stress granules — pure temporal-axis case

Stress granules (SGs) are membraneless condensates that form in response to cellular stress, containing mRNA, RNA-binding proteins, and translation initiation factors. Under physiological conditions they assemble rapidly, sequester translationally stalled mRNAs during stress, and disassemble upon stress resolution, typically on minute-to-hour timescales. The RNA-binding protein FUS (fused in sarcoma) partitions reversibly between the nucleoplasm and stress granules as part of its normal function in transcription regulation and DNA damage response.

Patel et al. (2015, *Cell* 162:1066, doi:10.1016/j.cell.2015.07.047) established that ALS-associated mutations in FUS drive an aberrant phase transition from liquid-like droplets to fibrillar aggregates over time. Wild-type FUS forms dynamic droplets that exchange rapidly with the surrounding cytoplasm; mutant FUS forms droplets that progressively lose dynamic exchange and mature into persistent fibrillar assemblies. Critically, the conformational state of FUS in its liquid-like and fibrillar phases differs in dwell time rather than in its position within the cell: FUS is still in the SG compartment in both cases. What changes is τ_R .

Wolozin and Ivanov (2019, *Nat Rev Neurosci* 20:649, doi:10.1038/s41583-019-0222-5) consolidated the evidence that “stress granules should be transient; in neurodegeneration they become chronic persistent.” Vanderweyde et al. (2012, *J Neurosci* 32:8270, doi:10.1523/JNEUROSCI.1592-12.2012) had already established that transient SG formation is necessary for cellular function, while hyperactive or prolonged SG formation leads to persistent SGs that nucleate pathological aggregation of tau, FUS, TDP-43, and other proteins.

The most direct experimental demonstration of the framework’s claim is Zhang et al. (2019, *eLife* 8:e39578, doi:10.7554/eLife.39578), which used optogenetic tools to independently control whether SGs formed and how long they persisted, in the absence of any cellular stressor. By decoupling SG assembly from its normal upstream triggers, the authors showed that chronic or chronic-intermittent SG assembly is *intrinsically* cytotoxic — meaning the pathology arises from the temporal residence pattern of the condensate itself, not from the stress that would normally trigger it. The SG substrate is unchanged. FUS and the other SG constituents are in their normal compartment. Only τ_R is pathological.

For the framework’s observable: in this case $R(s, c) = \tau_R(s_{SG}) \times \pi_c(\text{SG-compartment})$ reduces approximately to τ_R alone, because π_c is held constant by the biology. Pathology corresponds to $\tau_R(s_{SG})$ escaping its design-intent bounds — dwell time in the SG state persisting well past the timescale at which normal stress resolution would have disassembled the condensate. Recovery-path machinery at this substrate is the granulostasis system described by Alberti et al. (2017, *Front Mol Neurosci* 10:84, doi:10.3389/fnmol.2017.00084) — the HSPB8-BAG3-HSP70 axis that enforces SG disassembly and is the only explicitly named non-residence enforcement subsystem in the proteostasis literature. Paper 14 §6 argues that the full

proteostasis machinery (HSR/UPR/ISR, chaperone cycles, UPS, autophagy) should be understood as the broader class of which granulostasis is one explicit instance.

This is the cleanest *temporal*-axis case in the proteostasis literature. Pytel & Fromm Longo (2025) state that proteostasis governs “timing, location, and stoichiometry.” The FUS-in-SG case isolates timing, holding location approximately constant.

3.2 PrP topology and membrane anchoring — pure spatial-axis case

The prion protein PrP is a glycosylphosphatidylinositol (GPI) anchored cell-surface protein with multiple alternative topologies that are normally suppressed. The PrP^C native form is attached to the outer leaflet of the plasma membrane via its GPI anchor, with its polypeptide chain facing the extracellular space. Under mutation or stress, alternative topologies become populated: ^NPrP (N-terminus in the cytosol, C-terminus extracellular), ^CPrP (C-terminus in cytosol), cytosolic PrP (no membrane association), and the scrapie form PrP^{Sc} which may remain anchored or become released.

The spatial-axis case has the cleanest causal-localization experiments in all of neurodegeneration. Hegde et al. (1998, *Science* 279:827, PMID 9452375) showed that mutations shifting PrP to the transmembrane ^CPrP topology produce disease even when the protein’s conformational state is otherwise unremarkable. The pathology is spatial: the same peptide chain, differently positioned with respect to the membrane, produces different phenotypes.

Chesebro et al. (2005, *Science* 308:1435, PMID 15933194) ran the single most decisive experiment in the field: they generated transgenic mice expressing PrP with no GPI anchor (anchorless PrP, which cannot attach to the plasma membrane). After infection with prion strains, these mice accumulated large amounts of PrP^{Sc} amyloid deposits — substantially more than controls — but did not develop classical scrapie disease. **The deposition and the toxicity dissociated completely.** Anchorless PrP that would be pathogenic in the membrane-resident configuration was non-pathogenic (though still amyloidogenic) when its spatial residence was forced away from the plasma membrane. This is the canonical demonstration that *location, not presence*, is the disease-determining variable.

Subsequent work refined the specificity of the claim. Solomon et al. (2011, *J Biol Chem* 286:14724, doi:10.1074/jbc.M110.214973) established that mutant PrP toxicity requires plasma-membrane localization: redirecting mutant PrP away from the plasma membrane abolished its toxicity. Fehlinger et al. (2017, *Sci Rep* 7:7756, doi:10.1038/s41598-017-07260-2) showed that different prion strains use different endocytic routing, and Gatdula et al. (2026, *PLOS Pathogens* 22:e1013911, doi:10.1371/journal.ppat.1013911) demonstrated that membrane- anchored PrP^{Sc} is the proximate trigger of synaptotoxicity, rather than PrP^{Sc} released into the extracellular space. Across these studies, the consistent principle is that PrP pathology is a function of spatial residence: which compartment, with what orientation, and with what membrane association.

For the framework’s observable: in this case $R(s, c) = \tau_R(s) \times \pi_c(\text{compartment})$ reduces approximately to π_c alone, because τ_R is held approximately constant (PrP folding kinetics are not the variable of interest across the topology studies). Pathology corresponds to π_c escaping its design-intent bounds — occupancy shifted from “GPI-anchored on outer plasma-membrane leaflet” to any of several alternative configurations (transmembrane, cytosolic, non-anchored amyloid in tissue).

This is the cleanest *spatial*-axis case. The Chesebro 2005 experiment is a lighthouse for the framework's claim because it directly proves that deposition of even large amounts of pathological material is insufficient for disease; what is required is residence in the pathogenic compartment. Single-axis temporal observables would treat anchored and anchorless PrP^{Sc} as “the same state with the same dwell time” and would predict the same pathology. The experimental result is that they produce qualitatively different outcomes. The two-axis observable predicts this.

3.3 Huntingtin nuclear versus mitochondrial residence — both-axes case

Huntington's disease (HD) is caused by an expanded CAG repeat in the huntingtin gene (*HTT*), producing mutant huntingtin (mHtt) protein with a polyglutamine tract longer than ~35 residues. The same mutant protein — same sequence, same conformational ensemble at the molecular level — produces two partially independent toxic phenotypes that cannot be described by a single axis of the framework's observable.

The nuclear-residence axis. Saudou et al. (1998, *Cell* 95:55, PMID 9778247) showed that mutant huntingtin induces apoptosis in cultured neurons, and that this toxicity depends on nuclear localization. Peters et al. (1999, *Mol Cell Neurosci* 14:121, PMID 10532806) extended this by appending heterologous nuclear localization signals (NLS) or nuclear export signals (NES) to mHtt in cultured cells. Adding an NLS, which drives mHtt into the nucleus, *increased* toxicity. Adding an NES, which excludes mHtt from the nucleus, *decreased* toxicity. This was direct causal-localization manipulation: the same protein, made to reside in different compartments, produced different disease severity. The finding has been replicated in multiple in vivo systems; it is a settled result.

The mitochondrial-residence axis. More recent work (Yablonska et al. 2025, PMID 39779371; and earlier literature on mHtt- mitochondrial interactions through the early 2020s) has established that mutant huntingtin also associates with the mitochondrial outer membrane and interferes with mitochondrial dynamics, respiratory function, and apoptosis signaling. Yablonska et al. specifically showed that N17-domain phosphorylation regulates whether mHtt targets mitochondria, and that reducing mitochondrial mHtt residence ameliorates toxic phenotypes independent of changes to nuclear residence. This is a second, partially independent spatial-residence axis of pathology.

The temporal-residence axis is also non-trivial. The polyglutamine tract drives a slow conformational aggregation pathway with oligomeric intermediates, protofibrils, and mature fibrils. Which kinetic species dominates, and how long it dwells, varies with CAG-repeat length, cellular context, and chaperone engagement. The Wolynes-school aggregation-funnel literature (Thirumalai, Schuler, Wetzel, and successors) has characterized this dwell- time landscape extensively. The phenotypic consequences of different dwell-time regimes are active research.

This is a both-axes case. Single-axis observables cannot distinguish nuclear mHtt pathology from mitochondrial mHtt pathology, even though the experimental literature has established they are partially independent. The same protein, the same aggregation kinetics, but different compartmental residence — different disease. The two-axis observable $R(s, c)$ predicts and captures this: the s axis captures the conformational-kinetic contribution, and the c axis captures the nuclear-vs- mitochondrial contribution. Both are needed. A therapeutic strategy

that abolishes nuclear mHtt residence (e.g., via NES- conjugated degraders) would address one axis; a strategy that abolishes mitochondrial mHtt residence (e.g., via N17- phosphorylation manipulation) would address the other. The framework predicts that maximum therapeutic effect requires both.

3.4 Tau multi-compartment mislocalization — both-axes case

The microtubule-associated protein tau is normally an axonal protein that stabilizes microtubules in neurons. In tauopathies (Alzheimer's disease, frontotemporal dementia, progressive supranuclear palsy, and others), tau undergoes multiple pathological transformations: hyperphosphorylation, conformational changes, oligomerization, fibrillization, and — critically for the framework — mislocalization from axons to at least three distinct alternative compartments, each associated with different toxic phenotypes. This is the most complex of the four cases and most clearly requires the joint two-axis observable.

The dendritic spine axis. Ittner et al. (2010, *Cell* 142:387, doi:10.1016/j.cell.2010.06.036) demonstrated that tau mediates amyloid- β toxicity specifically through its mislocalization to dendritic spines, where it facilitates postsynaptic signaling dysregulation via Fyn kinase. Tau-knockout mice are protected from A β -induced toxicity, and the protective effect is rescued by re-expression of tau but not by re-expression of a tau mutant that cannot access dendritic spines. Hoover et al. (2010, *Neuron* 68:1067, PMID 21172610) showed that tau mislocalization to dendritic spines directly causes synaptic dysfunction, and that this mislocalization precedes overt tau aggregation. The pathology is spatial: tau in the axon is functional; tau in the dendritic spine compartment is pathogenic.

The RNP / condensate axis. Ash et al. (2021, *PNAS* 118: e2014188118, doi:10.1073/pnas.2014188118) and Jiang et al. (2019, *Acta Neuropathol Commun* 7:72) established that tau participates in liquid-liquid phase separation together with TIA-1 and other stress-granule-associated proteins, and that this condensate formation potentiates tau aggregation. The RNP-tau condensate is a qualitatively different compartment from both axonal and dendritic-spine tau, with distinct assembly dynamics and distinct toxic mechanisms. At this compartment, temporal residence (how long tau remains in the condensate before it either disassembles back to soluble tau or matures into fibrils) is the dominant variable.

The nuclear axis. Lester et al. (2021, *Neuron* 109:1675, PMID 33848474) showed that tau aggregates mislocalize nuclear speckle components and disrupt nuclear pore complex function. More recently, Yuan et al. (2026, *Acta Neuropathol* 147:113, doi:10.1007/s00401-026-02979-7) demonstrated that oligomeric tau produces nuclear lamina invagination, disrupting nuclear architecture. This is a third qualitatively distinct spatial residence regime for tau, with yet another set of toxic mechanisms.

Tau therefore produces at least three qualitatively distinct toxic phenotypes corresponding to three different compartmental residence regimes, plus a fourth temporal-residence contribution from RNP-condensate dynamics. No single-axis observable can capture this. A purely temporal observable — treating tau as a conformational ensemble with some dwell-time distribution — collapses dendritic-spine, RNP-condensate, and nuclear tau into one pooled measurement and cannot distinguish their therapeutic

implications. A purely spatial observable — treating tau as a compartment-occupancy measurement — misses the RNP-condensate temporal-dwell contribution entirely, because the RNP condensate is a transient assembly whose pathology depends on residence time, not just occupancy probability.

The two-axis observable $R(s, c)$ handles this case directly. Each of the three spatial regimes contributes a distinct $\pi_c(c_i)$ term ($c_{\text{dendritic}}$, c_{RNP} , c_{nuclear}), and the RNP-condensate regime contributes a temporal term $\tau_R(s_{\text{condensate}})$ that is pathology-critical independent of which compartment is occupied. The framework’s therapeutic prediction is circuit-specific: an intervention that reduces dendritic-spine tau will not help with nuclear-envelope pathology, and vice versa; maximum therapeutic efficacy in a given patient requires matching the intervention to the patient’s dominant residence-regime signature. This is consistent with clinical heterogeneity across tauopathies and suggests a diagnostic stratification strategy based on residence-regime profiling rather than on total tau burden.

3.5 Summary: two axes, four cases, one framework

The four cases jointly demonstrate what the framework’s two-axis observable captures that single-axis approaches miss:

Case	τ_R contribution	π_c contribution	Single-axis sufficient?
3.1 FUS in stress granules	Dominant (dwell past disassembly)	Constant (SG compartment)	Yes, temporal-only
3.2 PrP topology	Constant	Dominant (membrane-anchor configuration)	Yes, spatial-only
3.3 mHtt nuclear vs mitochondrial	Variable (aggregation kinetics)	Partitioned (nucleus vs mitochondria)	No
3.4 Tau multi-compartment	Variable (RNP-condensate dynamics)	Partitioned (dendritic, RNP, nuclear)	No

Cases 3.1 and 3.2 are “pure” in the sense that one axis is approximately held constant by the biology, which allows single-axis observables to describe them. Cases 3.3 and 3.4 have both axes actively contributing, and single-axis observables *necessarily* collapse distinctions that the experimental literature has established are real and therapeutically relevant. $R(s, c) = \tau_R(s) \times \pi_c$ is the minimal formalism that covers all four.

The cross-class observation that emerges from these cases, which we develop in the next section, is that the same two-axis residence-pathology structure recurs across at least the proteostatic substrate, with the expectation that it generalizes to substrates where relation-first ontology has been independently established (see §8 for the cross-substrate demonstration).

4. The Cross-Class Unifying Claim

The four canonical cases in §3 cover four distinct disease classes: condensate-pathology (FUS in stress granules), prion- class topology pathology (PrP), polyglutamine expansion disorder (huntingtin), and tauopathy. These classes have historically been studied separately, with their own conferences, their own model systems, their own therapeutic-target lists, and their own communities of investigators. The observation we develop in this section is that all four are coordinate instances of one structural principle: pathology in each case is the residence pattern of states and compartments whose transient occupancy was the biological design intent, rather than the identity of the states or compartments themselves.

4.1 Three mechanistic rules recur

Reading the spatial-residence cases across all four disease classes, three mechanistic rules recur, each of which has established primary literature in its own subfield and which together constitute the principle the framework names. These rules are descriptive observations, not novel claims; we name them here to make explicit that they recur across the four classes.

Rule 1 — Residence sets the local physicochemical regime. Acidic endosomes drive amyloid-precursor-protein β -cleavage differently from neutral cytoplasm (Das et al. 2013); ER topology gates which PrP topomers can form (Hegde et al. 1998); RNP- condensate chemistry potentiates tau oligomerization (Ash et al. 2021); the mitochondrial outer-membrane environment supports different mHtt toxic interactions than the nucleus (Yablonska et al. 2025). The protein's residence determines what physicochemistry is acting on it, and that physicochemistry shapes which states are populated and which interactions are accessible.

Rule 2 — Residence sets the accessible interactome and signaling space. Postsynaptic tau accesses Fyn kinase signaling that axonal tau cannot (Ittner et al. 2010); surface PrP accesses prion- specific neurotoxic signaling pathways that cytosolic or anchorless PrP cannot (Solomon et al. 2011); nuclear huntingtin accesses chromatin and transcription factors that cytoplasmic huntingtin cannot (Saudou et al. 1998; Peters et al. 1999); stress granule- resident proteins access translation-regulation machinery that soluble nuclear or cytoplasmic versions of the same proteins do not. The protein's residence determines what other molecular partners it can engage, and that interactome shapes which signaling consequences are possible.

Rule 3 — Residence selects the operative quality-control or return-path-machinery node. ER-resident misfolded proteins are subject to ER-associated degradation (ERAD); endolysosomal proteins are subject to endolysosomal sorting and degradation; cytosolic and nuclear proteins are subject to the cytosolic- nuclear chaperone-proteasome system and selective autophagy; organelle-resident proteins are subject to organelle-specific proteostasis (mitochondrial PQC, peroxisomal PQC, and so on); SG-resident proteins are subject to granulostasis (Alberti et al. 2017). The protein's compartmental residence determines which proteostasis network is responsible for clearing or refolding it, and failure of that network is what converts transient residence into persistent pathological residence.

4.2 The unifying claim

These three rules, taken together with the two-axis observable $R(s, c) = \tau_R(s) \times \pi_c$, support the framework's central claim:

Pathology in any system with both regimes available to it is the temporal-and-spatial residence pattern of states whose transient sampling was the design intent, produced by failure of the return-path machinery that would have enforced return to functional control, not the identity of the states themselves.

Three points warrant emphasis.

First, this is a claim about residence patterns rather than about states per se. In each of the four cases in §3, the pathological state is also visited by healthy systems: FUS forms transient SG droplets normally; PrP populates alternative topologies as part of normal quality-control surveillance; huntingtin shuttles to the nucleus and engages mitochondria as part of normal function; tau visits dendritic compartments at low frequency under normal conditions. Pathology emerges not when these states are visited but when residence in them escapes design-intent bounds.

Second, this is a claim about return-path machinery rather than about the pathological state itself. The framework predicts that therapeutic strategies targeting the *state* will be partially effective (because reducing the rate of pathological-state formation reduces the integrated residence-time accumulation) but that therapeutic strategies restoring the *return-path machinery* will be more durably effective (because they re-enable the physiological mechanism that converts residence into transit). This is testable; we propose it as Experiment 7 in §11.

Third, this is a claim about substrate-general principle rather than about proteostasis specifically. The four cases in §3 are drawn from proteostasis because the proteostasis literature has the most complete causal-localization experiments; but the framework predicts the same residence-pathology structure in any substrate where (a) both regimes are available to the system, (b) explicit return-path machinery exists, and (c) recovery kinetics can be measured. §8 demonstrates this for receptor pharmacology, cancer phospho-regulation, autonomic physiology, exercise physiology, large-scale brain network dynamics, intersubjective coupling, phenomenology, and recursive AI/data dynamics.

4.3 What this is not

The unifying claim is structurally narrow even though it spans multiple substrates. We are not claiming that all disease is residence pathology. There are pathologies of state-identity (a protein that is genuinely absent because of a null mutation; a receptor that has been deleted; an irreversible structural damage); these are not residence pathologies, and the framework makes no claim about them. We are not claiming that the residence-pathology principle predicts which specific intervention will work in any specific patient (clinical heterogeneity makes this a complex stratification problem, not a single-formula problem). And we are not claiming that the principle reduces to a single biological mechanism (the principle is substrate- uniform but its molecular implementations are circuit-specific in the sense discussed in §6).

What we are claiming is that an empirically tractable two-axis observable, a substrate-uniform residence-pathology grammar, and a cross-substrate demonstration of the principle have not been assembled together before, and that doing so opens an integrated program of diagnostic, prognostic, and therapeutic work across substrates that the current substrate-specific literatures cannot support individually.

See Figure 4 for the visual summary of the five symmetric strategies introduced below.

5. Symmetric Formulation of Kinetic Stabilization

The cleanest existing therapeutic strategy in the proteostasis literature is **kinetic stabilization**, developed by Kelly and colleagues for transthyretin amyloidosis. Bulawa, Kelly et al. (2012, *PNAS* 109:9629, doi:10.1073/pnas.1121005109) introduced tafamidis, a small molecule that binds the transthyretin tetramer at its T4-binding sites and stabilizes the native tetrameric state, slowing its dissociation into monomers that would otherwise misfold and aggregate. The therapeutic mechanism is to *extend the dwell time of the native state*: increase $\tau_R(s_{\text{native}})$ so that the equilibrium populates s_{native} rather than the pathological aggregation pathway. Tafamidis has been approved for clinical use in multiple transthyretin amyloid syndromes and is the paradigmatic kinetic-stabilization drug.

Kelly's formulation is *asymmetric*. The therapeutic move is to increase residence in the native state; the pathological state is implicitly treated as the absence of native residence rather than as a residence-bearing object in its own right. This asymmetry is appropriate to the transthyretin biology, where the native tetramer and the misfolded monomer are clearly different states with clearly different dwell-time targets. But the framework's two-axis observable suggests that kinetic stabilization is one special case of a more general therapeutic family.

5.1 The symmetric formulation

If pathology is residence pattern rather than state identity, then therapeutic strategies that alter the residence pattern can operate in any of three directions, each of which targets a different term in $R(s, c) = \tau_R(s) \times \pi_c$:

Strategy A — Extend native temporal residence (Kelly). Stabilize $\tau_R(s_{\text{native}})$. Tafamidis is the canonical example. The therapeutic target is the native conformational state, and the intervention increases its dwell time so the equilibrium favors function.

Strategy B — Destabilize pathological temporal residence (symmetric inverse of Kelly). Reduce $\tau_R(s_{\text{pathological}})$. Interventions in this class include conformation-selective degraders, immunotherapies targeting pathological conformations (such as the antibodies developed against amyloid- β and tau oligomeric species), and pharmacological chaperones that destabilize aggregation-prone intermediates. The therapeutic logic is symmetric to Strategy A: rather than holding the system in the functional state, reduce the dwell time in the pathological state so the system returns to function more rapidly.

Strategy C — Route protein away from pathological spatial residence. Reduce $\pi_c(c_{\text{pathological}})$. Examples include NES-tagged constructs that exclude mutant huntingtin from the nucleus (Peters et al. 1999 directly demonstrates this in cell models; the principle extends to NES-conjugated bifunctional degraders in development); GPI-modification approaches that alter prion subcellular localization (theoretical, following Chesebro 2005); compartment-selective autophagy inducers that preferentially clear protein from one compartment while sparing function in another. The therapeutic logic is on the spatial axis rather than the temporal axis: the state may persist, but its residence in the pathological compartment is abolished or reduced.

Strategy D — Route protein toward protective spatial residence. Increase $\pi_c(\text{c_protective})$. The symmetric inverse of Strategy C on the spatial axis. Examples include enhanced lysosomal targeting in diseases where lysosomal residence permits clearance, or selective autophagy-receptor engineering that directs pathological conformations to autophagosomal compartments.

Strategy E — Restore return-path machinery. Increase the endogenous rate at which the system transitions from pathological residence back to functional residence, on either axis. Examples include HSR/UPR amplification (the arimoclomol class, with translational caveats discussed below), chaperone-protein gene therapy, autophagy enhancers, and immunomodulatory approaches that restore granulostasis or PQC system function. The therapeutic logic targets the recovery machinery itself rather than any particular state or compartment; the framework predicts this will be the most durably effective strategy class but also the hardest to engineer, since the machinery operates at multiple scales simultaneously.

5.2 Differentiation from Knowles-Vendruscolo-Dobson

The most adjacent existing framework is Knowles, Vendruscolo and Dobson's *protein metastasis* (Knowles, Vendruscolo & Dobson 2014, *Nat Rev Mol Cell Biol* 15:384, doi:10.1038/nrm3810). In their formulation, the native state is a kinetically metastable basin, and the amyloid state is the thermodynamic ground state; proteins live functionally in the metastable basin and occasionally cross the barrier into the pathological basin, with proteostasis machinery returning them. The framework presented here differs in three ways.

First, Knowles-Vendruscolo-Dobson treats the native and pathological basins as different states. The framework here allows that the *same* state can be functional or pathological depending on its residence pattern: this is what §3.1 (FUS in SGs) and §3.4 (tau in different compartments) demonstrate. The FUS in a healthy transient SG and the FUS in a chronic pathological SG are not different states; they are the same state with different τ_R .

Second, the protein-metastasis framework is implicitly asymmetric (native is metastable visit, amyloid is thermodynamic residence; the therapeutic strategy is to deepen the metastable basin against the pathological one). The framework here is symmetric: residence-pathology can be alleviated by extending functional residence, abolishing pathological residence, or restoring the transit machinery, in any combination.

Third, protein metastasis is substrate-specific (it is a theory about proteins). The framework here predicts the same structural principle in receptor pharmacology, autonomic regulation, brain- network dynamics, and AI/data substrates (see §8). Protein metastasis is one instance of the principle the framework names.

5.3 Translational caution

The framework's prediction that return-path-machinery restoration is the most durable therapeutic strategy is *consistent with*, but not *guaranteed by*, the principle. A central translational caution is the arimoclomol case: arimoclomol is a heat-shock- response amplifier with substantial preclinical data supporting proteostasis enhancement, and it has been developed as a candidate disease-modifying therapy for amyotrophic lateral sclerosis. The ORARIALS-01 phase 3 trial (results reported 2024) was negative on its primary endpoint despite

the mechanistic rationale. The framework treats this as a *cautionary instance* rather than a refutation: identifying the principle correctly does not guarantee that any specific intervention based on it will work, because target engagement, timing, dose, leverage, and patient stratification all matter independently. The principle says where to look. Whether a given molecule, dose, or schedule finds the right point in the residence-pattern landscape is an empirical question that the principle does not answer by itself.

See Figure 3 for the four-panel typology of return-path machinery classes elaborated below, with their shared exit-rate observable $\lambda_{\text{exit}} = 1/\tau_{\text{R}}(s,c)$.

6. Return-Path Machinery as a Multiscale Class

The proteostasis literature has independently developed an extensive vocabulary for what we are calling *return-path machinery*: HSP70 ATP-dependent client cycling, HSP90 client-dwell modulation, ubiquitin-proteasome system (UPS) clearance, selective and bulk autophagy, granulostasis (HSPB8-BAG3-HSP70 for stress-granule disassembly), the heat-shock response (HSR), unfolded protein response (UPR), integrated stress response (ISR), endoplasmic-reticulum-associated degradation (ERAD), endolysosomal sorting, NLS/NES-mediated nucleocytoplasmic transport machinery, nuclear pore complex regulation, and organelle-specific quality-control systems including mitochondrial and peroxisomal PQC. These have historically been treated as a heterogeneous toolkit — each with its own substrate specificity, its own regulatory logic, its own community of investigators.

We propose that this toolkit, plus three further substrate-specific classes, constitutes a single observable category: **non-residence enforcement machinery**, characterized by a common observable (recovery time constant τ_{recovery} , or equivalently the dominant relaxation rate of the system's return from perturbation) and a shared structural role (preventing pathological residence on either axis of $R(s, c)$).

6.1 Four substrate-specific machinery classes

Class 1: Proteostatic return-path machinery. The proteostasis literature listed above. The unit of operation is the protein, and the machinery enforces transit between conformational states and between subcellular compartments. Failure produces aggregation- pathology (Strategy A-E from §5 target this machinery either symmetrically or asymmetrically).

Class 2: Phospho-regulatory return-path machinery. Protein phosphatase 2A (PP2A) and other serine/threonine phosphatases function as dephosphorylation resetters that return signaling proteins from activated to baseline states; the APC/Axin/GSK3 β destruction-complex logic enforces protein turnover via phospho-dependent ubiquitination; the broader class of E3 ligases acts as substrate-turnover machinery. Failure of this class produces oncogenic-signaling residence (CIP2A inhibition of PP2A in leukemia, for instance, or persistent AKT/MAPK activation in tumors).

Class 3: Autonomic return-path machinery. The vagal brake and baroreflex re-engagement after sympathetic challenge are the best-characterized autonomic return mechanisms. The cardiac vagal response after acute stress (Souza et al. 2007, Framingham Heart Study HRV data) provides a clean observable: reduced HRV predicts mortality, and the temporal pattern of vagal

recovery after a stressor is the direct individual-level recovery-time- constant analog at this substrate.

Class 4: Network-dynamical return-path machinery. The salience network and central executive network mediate switching from internal-mode states back to task-positive cognition; task- induced deactivation of the default-mode network is the observable signature of this return; effective connectivity from CEN to DMN under cognitive demand is the directional indicator. Failure of this class produces return-to-task-set failure, which is what the canalization-of- psychopathology literature names at the brain substrate (Carhart-Harris, Chandaria, Friston et al. 2023).

6.2 The common observable

Across all four classes, the same family of observables applies:

- recovery time constant τ_{recovery} ,
- relaxation rate $\alpha = 1/\tau_{\text{recovery}}$,
- dominant restoring eigenvalue $\text{Re}(\lambda_{\text{max}})$ of the linearized return dynamics,
- dwell-time persistence on the pathological-residence side,
- substrate-specific instantiations: aggregate clearance flux, baroreflex gain, phospho-state settling time, default-mode dwell time.

This vocabulary is already developed in the engineering-resilience and critical-slowness-down literatures (van Nes & Scheffer 2007, *Am Nat* 169:738, doi:10.1086/516845; van de Leemput et al. 2014, *PNAS* 111:87, doi:10.1073/pnas.1312114110) and in the loss-of- complexity framework (Lipsitz & Goldberger 1992, *JAMA* 267:1806). These sources establish the mathematical machinery; the framework's contribution is *cross-substrate uniformity* — the same observable, measured at substrate-appropriate timescales, applied to all four machinery classes.

6.3 Circuit-specificity discipline

A central discipline of the framework is that return-path framing must be **circuit-specific**, not universal. The same regulator plays the return-path role in one circuit and the pathological-residence-enforcement role in another. GSK3 β is the standard example: in the Wnt/ β -catenin pathway, GSK3 β phosphorylates β -catenin for degradation, enforcing transit out of the pathological- residence state; in the NF- κ B / mitosis / survival pathway, GSK3 β supports oncogenic signaling. PP2A is similarly context-dependent; the DT-061 and iHAP1 small-molecule PP2A activators have shown mechanism-disputed effects (some reported activity is PP2A- independent in subsequent assays).

The framework therefore specifies that pathology is failure of the *circuit-specific* return-path operator, not loss of any one molecular activity in isolation. A therapeutic strategy that restores PP2A activity in a tumor where PP2A acts as return-path machinery (CIP2A-positive leukemias, for instance) is a residence-restoration intervention; the same molecule used in a context where PP2A plays a different role could be ineffective or counterproductive. Circuit-level diagnosis precedes machinery- restoration therapy.

6.4 Allostasis caveat

A second discipline is the *allostatic* caveat (McEwen 1998, *Ann NY Acad Sci* 840:33, doi:10.1111/j.1749-6632.1998.tb09546.x). Return-path machinery does not always return a scalar variable to one invariant baseline; healthy systems often shift operating points adaptively. The framework defines return-path failure as inability to restore *functional controllability* rather than as deviation from a specific prior value. A system that adaptively shifts its set-point in response to chronic input change is functioning, not failing, as long as it can still return to operational range after further perturbation. Pathology is the loss of the restoration capacity, not the shifted set-point.

This caveat matters operationally because the framework's recovery-time-constant observable can in principle be measured relative to baseline or relative to a moving target. The appropriate referent is the moving target — the current functional control manifold — not a fixed historical baseline. This is consistent with the FEP / active-inference treatment of self-organization (Friston, Levin, Sengupta & Pezzulo 2015, doi:10.1098/rsif.2014.1383; Ramstead, Badcock & Friston 2018, doi:10.1016/j.plrev.2017.09.001), which the framework treats as formal scaffolding for the recovery-kinetics observable even though, as discussed in §1, the FEP literature stops short of the explicit ontological move the framework makes.

7. Frustration Without Escape

Wolynes and Ferreiro have developed an extensive theory of *local frustration* in protein folding (Ferreiro et al. 2007, *Proc Natl Acad Sci USA* 104:19819; the recent 2025 review Parra, Komives, Wolynes & Ferreiro provides a current summary). The framework's central observation is that *frustrated* interactions are local energetic conflicts that the protein cannot fully satisfy by folding into its native state; rather than being defects, these frustrated regions are functionally important because they are often the sites of conformational flexibility, ligand binding, allosteric regulation, and protein-protein interaction interfaces.

Frustration theory is *state-shaped*. It describes which interactions are frustrated in a given conformational state, with the metric of frustration computed from local energetic conflicts between residues. The theory is formally developed and has clinical relevance: disease-associated mutations frequently occur at frustrated sites, and the patterns of frustration distinguish functional ensembles from pathological ones.

7.1 Extension to residence-shaped frustration

The framework proposes a complementary extension: *frustration without escape*. Frustration theory describes the existence of local energetic conflict in a state; the framework's extension describes how long the system *resides* in the frustrated configuration before the conflict is resolved.

In the healthy regime, a protein visits frustrated configurations transiently as part of normal function (catalysis, allosteric response, binding-partner exchange). The frustrated configuration is a transit state with bounded τ_R . The return-path machinery (chaperones, allosteric communication, conformational flexibility) resolves the frustration on functional timescales.

In the pathological regime, frustration without escape:

- the system visits the same frustrated configurations, but $\tau_R(s_{\text{frustrated}})$ escapes its design-intent bounds;
- the return-path machinery that would have resolved the frustration is impaired or absent;
- the prolonged residence in the frustrated state allows alternative pathways to populate (oligomerization, aberrant protein-protein interactions, allosteric capture by pathological partners).

The pathology is not the frustration itself — frustration is physiological. The pathology is the failure to escape it.

7.2 Position relative to the published frustration literature

“Frustration without escape” as an explicit framework concept is not found in the published frustration theory literature at the time of writing. We do not claim that Wolynes-Ferreiro collaborators are unaware that frustration must be resolved on functional timescales — this is implicit in the theory’s functional framing. What is not made explicit is the residence-time treatment of frustration as a separate analytical level: an “escape rate from frustration” or “frustration-dwell- time distribution” as a measurable quantity that distinguishes healthy from pathological proteins beyond the static frustration metric.

The framework’s extension is therefore a refinement of frustration theory rather than a competitor. We propose that frustration-dwell-time profiles are diagnostic for residence-pathology in the proteostasis substrate, and that the same structural principle (transit-state residence becoming pathological residence) operates across the cross-substrate cases discussed in §8. Pre-submission, we will circulate a courtesy preprint to the Wolynes and Ferreiro laboratories so that the relationship between frustration theory and the framework’s extension can be calibrated by the principal authors of the underlying theory.

7.3 Why this matters for the framework

Frustration without escape is the framework’s most precise statement of the residence-pathology principle at the proteostasis substrate, because it gives an exact mechanistic content to “residence past design intent.” The design intent is the functional transit time across the frustrated configuration; the pathological extension is the integrated residence in the same configuration. The metric is concrete: dwell time on frustrated residues, measured by molecular-dynamics simulations or by experimental approaches such as hydrogen-deuterium exchange. Where existing frustration theory characterizes the static map of local conflicts in a protein, the framework’s extension provides the dynamical observable that distinguishes healthy from pathological dynamics on that map.

8. Cross-Substrate Extensions

The framework’s central claim is that residence-pathology is a substrate-general principle: where the four canonical disease cases in §3 sit within the proteostasis substrate, the same two-axis observable $R(s, c) = \tau_R(s) \times \pi_c$ and the same return- path-machinery structure apply wherever a system has (a) both regimes available, (b) explicit return-path machinery, and (c)

measurable recovery kinetics. This section demonstrates the claim across seven further substrates, each of which has independently developed its own residence-pattern vocabulary without, to our knowledge, being united under a single formalism.

The substrates are presented in order of decreasing biological scale: receptor pharmacology at the molecular scale (§8.1), cancer phospho-regulation at the cellular-signaling scale (§8.2), autonomic physiology at the organ-system scale (§8.3), exercise physiology at the whole-organism scale (§8.4), large-scale brain network dynamics (§8.5) with its psychedelic-restoration subcase (§8.5.1), PCI phenomenology at the first-person scale (§8.6), and the AI/data substrate at the human-technology coupling scale (§8.7). Each subsection follows a common structure: identify the substrate's residence-pattern observable, identify its return-path machinery, cite substrate-level prior art for the principle (which is often extensive but not cross-substrate), and note the specific way in which the framework's two-axis formalism extends what the substrate literature already possesses.

8.1 Pharmacology — the Copeland residence-time paradigm

Drug-receptor binding kinetics provides the cleanest single substrate for the residence-pathology principle because the visits-versus-residence distinction is already explicit in the literature. Tummino & Copeland (2008, *Biochemistry* 47:5481, doi:10.1021/bi8002023) established the framework of *residence time* $\tau = 1/k_{\text{off}}$ as the decision variable in drug discovery, rather than equilibrium affinity K_d . In an open system (such as the body, where drug concentration fluctuates with absorption, distribution, metabolism, and excretion), what matters for efficacy is how long the drug stays bound when free drug concentrations fall below dissociation levels, not how tightly it binds at equilibrium.

Bosma et al. (2017, *Front Pharmacol* 8:667, doi:10.3389/fphar.2017.00667) demonstrated this at H1 histamine receptors using a series of clinically used antihistamines. Their data show Spearman $\rho = -1.0$ between receptor-recovery time and k_{off} (perfect rank correlation with residence time), but only $\rho = -0.6$ between receptor-recovery time and equilibrium affinity. The framework reads this as direct experimental demonstration that $\tau_R(s_{\text{bound}})$ is the dominant observable on the pharmacology substrate, with the spatial axis π_c approximately constant (receptor compartment). Casarosa et al. (2009, *JPET* 330:660) and Sykes et al. (2017, *Nat Commun* 8:763) extend the demonstration to muscarinic M3 and dopamine D2 receptors respectively.

Vauquelin's rebinding work (Vauquelin & Van Liefde 2005; Vauquelin 2016) adds a second axis: for some drugs, micro-pharmacokinetic rebinding (re-association after dissociation within a diffusion-limited local neighborhood) matters as much as or more than the intrinsic off-rate. This introduces a spatial-residence component to receptor pharmacology — occupancy probability in the tissue compartment where rebinding is efficient — which maps directly onto the framework's π_c axis.

Return-path machinery on this substrate is receptor desensitization, internalization, and recycling, which enforces that receptor occupancy returns to baseline on physiological timescales. Pathology on this substrate is prolonged receptor engagement past design-intent bounds — either by drugs with pathologically long residence times, by receptors with impaired desensitization, or by persistent agonist exposure that overwhelms recovery capacity.

The pharmacology literature thus already has the visits-versus- residence distinction explicit, with residence-time observable, rebinding-mediated spatial contribution, and return-path machinery all named. What it does not have is the cross-substrate unification: the claim that the same formalism applies to protein aggregation, autonomic regulation, or brain-network dynamics. The framework's contribution at this substrate is citation and uniformity rather than novelty.

8.2 Cancer phospho-regulation

Cancer signaling provides a cellular-scale demonstration of the residence-pathology principle with well-characterized return-path machinery. Oncogenic signaling frequently involves *persistent phosphorylation* of substrates whose normal function requires transient phospho-activation followed by rapid dephosphorylation. The return-path machinery is the protein-phosphatase system, principally protein phosphatase 2A (PP2A), which acts as the dephosphorylation resetter for many oncogenic kinase substrates including Akt, MAPKs, c-Myc, and β -catenin.

PP2A is frequently inactivated in cancer, not by loss of expression but by *endogenous inhibitors* — CIP2A and SET — that bind the PP2A holoenzyme and sequester it from its substrates. CIP2A overexpression is observed in a majority of solid tumors and hematologic malignancies; its inhibition of PP2A produces persistent c-Myc, AKT, and ERK phosphorylation, locking the cancer cell in proliferative-signaling residence. SET similarly sequesters PP2A in leukemias. The therapeutic strategy of SET antagonism (with molecules such as OP449) or SMAP-class small-molecule activators of PP2A (SMAPs) aims to restore the dephosphorylation return path.

An orthogonal machinery node is the APC/Axin/GSK3 β destruction complex, which phosphorylates β -catenin to mark it for ubiquitin-mediated degradation. In Wnt-dependent cancers (colon cancer in particular), APC mutations destroy the destruction complex and prevent β -catenin return to baseline; β -catenin residences in the nuclear-signaling state and drives oncogenic transcription.

The framework reads both pathologies as return-path-machinery failure at the phospho-regulatory substrate: PP2A inhibition reduces the dephosphorylation rate for multiple kinase substrates; APC loss reduces the degradation rate for β -catenin. The resulting pathology is persistent residence in phosphorylated / stabilized states that were designed to be transient.

Circuit-specificity discipline applies strictly here. PP2A is a return-path operator in most oncogenic contexts, making its restoration anti-proliferative; but PP2A loss in other contexts has been associated with different pathologies, and the DT-061 and iHAP1 small-molecule activators have shown mechanism-disputed activity (some reported effects are PP2A-independent in subsequent assays). GSK3 β provides the clearest contextual reversal: in Wnt/ β -catenin signaling GSK3 β enforces transit out of pathological residence, while in NF- κ B / mitosis / survival signaling GSK3 β supports persistent activation. The framework predicts that therapeutic targeting must be circuit-matched to the return-path role in the patient's specific cancer.

8.3 Autonomic physiology — the vagal brake

Autonomic regulation of heart rate and blood pressure provides the best-characterized organ-system-scale instance of the return-path-machinery principle. The cardiovascular system is constantly perturbed by behavior, stress, exercise, postural change, and environmental demands; the vagal brake and baroreflex machinery enforces return to baseline on seconds-to-minutes timescales. Heart rate variability (HRV), particularly its high-frequency component that reflects parasympathetic (vagal) tone, is the direct individual-level recovery-time-constant analog for this substrate.

Framingham Heart Study data (Tsuiji et al. 1996, *Circulation* 94:2850) established that reduced HRV predicts all-cause and cardiac mortality in population cohorts. The temporal pattern of vagal recovery after acute stress (Souza et al. 2007; many later replications) provides a clean experimental handle: subjects perform a stressful task, heart rate elevates, and the time constant of recovery to baseline is a measurable property that varies across individuals and across health states. Slower recovery predicts worse outcomes across multiple cardiovascular and metabolic endpoints.

The framework reads this directly. The pathological regime is prolonged sympathetic-drive residence without adequate parasympathetic return. The return-path machinery is the vagal brake and baroreflex. The observable is the recovery time constant of cardiac autonomic tone after perturbation. Interventions that augment the return path — HRV biofeedback, vagal-nerve stimulation (VNS), controlled breathing practices, cold exposure — produce measurable improvements in the recovery observable and downstream clinical outcomes.

A clinical anchor for the cross-substrate claim comes from neutrophil immunology. CXCR4 is a chemokine receptor that mediates reverse-migration return of activated neutrophils from inflamed tissue back to bone marrow for clean apoptosis. JAM-C regulates neutrophil reverse transendothelial migration; the LTB₄-BLT1 axis controls directional swarming. Healthy inflammation resolution requires return-path machinery operating cleanly; in acute pancreatitis-associated lung injury, reduced JAM-C expression enables neutrophils to leak back into the circulation in pathological patterns that spread inflammation to distal organs. A 2022 *Nature Reviews Immunology* analysis frames reverse neutrophil migration as a “double-edged sword” — resolution when return is clean, pathology when return is corrupted. This is the framework’s boundary-KKT structure (Paper 12) translated directly to immune cells, with real disease, real mortality data, and named molecular machinery.

8.4 Exercise physiology — endogenous versus operator-sustained boundary operation

Exercise physiology provides a whole-organism instance of the residence-pathology principle with a specific distinction between healthy and pathological residence that the framework names precisely. Sustained high-intensity exercise places the organism at a physiological boundary against VO₂max, glycogen depletion, pH buffering limits, and thermoregulatory ceilings (Joyner & Coyle 2008, *J Physiol* 586:35–44; Burnley & Jones 2018, *Eur J Appl Physiol* 118:1–12). The healthy case is *endogenous* boundary operation with intact return-path machinery: phosphocreatine depletion, rising H⁺ from glycolysis, sympathetic-drive exhaustion, and metaboreceptor afferent signaling cumulatively enforce return from the boundary to a recovery regime, after which glycogen replenishes, pH buffers restore, and

parasympathetic tone re-engages (Coote 2010, *Exp Physiol* 95:431–440). The adaptations of athletic conditioning — improved mitochondrial density, expanded buffering capacity, enhanced vagal tone (HRV adaptations in trained athletes, Aubert, Seps & Beckers 2003, *Sports Med* 33:889–919) — are downstream of iterated visit-and-return, not of accumulated residence.

The pathological case is *operator-sustained* boundary operation in which pharmacological agents silence the return-path cutoffs. The empirical literature on supraphysiological-dose anabolic-androgenic steroid (AAS) use in strength athletes documents this pattern with particular clarity. Smit et al. (2021, *Frontiers in Reproductive Health*; the HAARLEM study) measured cardiac structure and function in amateur strength athletes before, during, and after a self-administered AAS cycle: left ventricular mass increased by 28.3 g (95% CI 14.2–42.4, $p < 0.001$), positively correlated with cumulative AAS dose, with impaired systolic and diastolic function during the operator-sustained period — and *all parameters returned to baseline at median 8-month recovery follow-up after AAS discontinuation*. The recent expert review (Liu, Wang & Zhang 2023, *Frontiers in Cardiovascular Medicine* 10:1233223) and the American College of Cardiology 2024 expert summary (Baggish et al. 2024) document the broader pattern: AAS-associated left ventricular hypertrophy with matrix collagen deposition and fibrosis, conferring elevated cardiomyopathy risk in chronic users. Parallel residence-pathology signatures appear in stimulant-cardiomyopathy case literature — amphetamine and methamphetamine cardiomyopathy with reduced LVEF that frequently recovers after cessation (Won et al. 2013, *Heart* 99:309–314; case reports including Patel et al. 2023, *Cureus* 15:e40719) — and in the ephedra/ephedrine adverse-event literature (Shekelle et al. 2003 *Annals of Internal Medicine* meta-analysis; American College of Cardiology summary 2003: 4 deaths, 5 myocardial infarctions, 11 cerebrovascular accidents, 4 seizures across reviewed adverse events, with approximately half occurring in users under 30 years of age).

The reversibility documented in the HAARLEM data is the framework's key residence-pathology signature: when the operator-sustained boundary residence ceases, the return-path machinery (in this case the endogenous androgen-receptor signaling restoration, anti-fibrotic remodeling, and vagal reactivation) reasserts, and the substrate-level pathology resolves. This is what distinguishes operator-sustained residence-pathology from substance-level structural damage — the structural changes are downstream of the residence pattern, not vice versa.

The two axes of $R(s, c)$ map onto this substrate cleanly. The temporal axis $\tau_R(s_{\text{boundary}})$ captures how long the organism resides at the physiological ceiling before return-machinery reasserts (operationally: weeks-to-months of AAS-cycle duration, hours-to-days of acute stimulant exposure, training-volume-times-stimulant-dose for compound exposures). The spatial axis π_c captures which compartments (cardiac, skeletal-muscle, hepatic, neuroendocrine) are over-occupied by the boundary-regime metabolic state, measured by compartment-specific imaging and functional assays. Health is visits with intact return (τ_R bounded by endogenous cutoffs); pathology is operator-sustained residence (τ_R pharmacologically extended past those cutoffs). The framework's prescription at this substrate — respect the endogenous cutoffs, cultivate the return path, refuse operator-sustained boundary residence — is consistent with the standard sports-medicine periodization literature (Bompa & Buzzichelli 2018, *Periodization*) and with the framework's cross-substrate residence-pathology principle. The HAARLEM reversibility data is the most direct empirical confirmation of the framework's principle at the exercise substrate currently available.

8.5 Large-scale brain network dynamics — return-to-task-set

Human psychopathology has developed the most explicit cross-disorder residence-pathology framing of any substrate outside proteostasis. Carhart-Harris, Chandaria, Friston et al. (2023, *Neuropharmacology* 226:109398, doi:10.1016/j.neuropharm.2022.109398) introduced the *canalization* framework, in which “cognitive and behavioral phenotypes that are regarded as psychopathological are canalized features of mind, brain, or behavior that have come to dominate an individual’s psychological state space.” They explicitly cross-apply this to depression, obsessive-compulsive disorder, addiction, post-traumatic stress disorder, eating disorders, psychosis, and somatoform conditions. Juliani, Safron & Kanai (2024, *Neurosci Conscious* niae005, doi:10.1093/nc/nae005) refine the framework into *Deep CANALs*, distinguishing Type-A inference-level canalization from Type-B synaptic-weight landscape canalization.

Canalization is direct brain-substrate prior art for the framework’s residence-pathology principle, and we cite it proximately (see §1). The framework extends canalization in three ways at this substrate.

First, we propose an **individual-level formal observable** $R(s, c) = \tau_R(s) \times \pi_c$ predictive of treatment outcome at the per-subject level. The closest existing observables are group-level: Singleton et al. (2022, *Nat Commun* 13:5812, doi:10.1038/s41467-022-33578-1) develops receptor-informed network control theory at the cross-individual correlation level; Vohryzek et al. (2024, *Brain Commun* 6:fcae049, doi:10.1093/braincomms/fcae049) fits group-averaged Hopf models for psilocybin responders and non-responders. Vidaurre’s HMM Fisher kernel (Vidaurre, Smith & Woolrich 2017, doi:10.1073/pnas.1705120114) is individual-level for cognitive traits but has not been applied to psychedelic treatment outcome. $R(s, c)$ with subject-specific dwell-time τ_R and compartmental (network-state) occupancy π_c fits as individual-level predictor.

Second, we sharpen the framing from **static DMN hyperconnectivity** to **return-to-task-set failure**. The depression literature does not actually support the simplified claim that depression is static default-mode-network hyperconnectivity; Wise et al. (2017) found DMN instability in depression rather than hyperstability; the REST-meta-MDD consortium found mixed and sometimes *reduced* within-DMN functional connectivity in recurrent MDD. What is consistent across the literature is *impaired suppression of DMN under task demand, prolonged dwell in internally-oriented network states, and reduced switching to task-positive network configurations*. This is return-path failure at the network-dynamical substrate: the observable is the time constant of transition from DMN-dominant to task-positive-dominant state following a cognitive demand cue, and pathology is prolonged $\tau_R(s_{\text{DMN}})$ past design-intent bounds.

Third, we integrate the Paper 12 v1.4 substrate-side demonstration of §3.5 smooth-interior and §3.6 boundary-KKT residence topology as the formal model of the canalization claim. This is discussed further in §8.6 (PCI phenomenology).

8.5.1 Psychedelic restoration of transitions

Psychedelic therapy provides a specific instance of return-path-machinery restoration at the network-dynamical substrate. The framework reads the therapeutic mechanism as the one-

sentence claim from the pharmacology-to-psychedelics synthesis (Lane 4 of the framework's prior-art surveys): "Copeland pharmacology teaches that efficacy can depend on how long a target remains occupied through biologically relevant intervals; psychedelic systems neuroscience increasingly teaches that therapy can depend on how easily the brain moves between recurrent states, not on whether it visits some wholly unprecedented region-defined state."

Lane 4 documented a five-link mechanism chain from 5-HT_{2A} receptor occupancy (molecular scale) to phenomenology (first-person scale), with established literature at each link: drug concentration → bound receptor fraction → residence pattern → downstream signaling timing → network-level gain and coupling → state-space geometry (dwell times, transition barriers, hierarchy flattening) → phenomenology and therapeutic outcome.

The explicit-restoration cluster of citations is load-bearing: Doss et al. (2021, psilocybin in MDD increased cognitive flexibility four-plus weeks post-treatment with increased ACC-PCC dynamic FC); Singleton et al. (2022) control-energy lowering for state transitions under LSD and psilocybin; Daws et al. (2022, psilocybin decreased modularity and increased global integration); Nardou et al. (2023, mouse: psychedelics reopened social-reward- learning critical period with oxytocin-LTD metaplastic restoration); Vohryzek et al. (2024, whole-brain modeling identifying regions mediating depressive-to-healthy transitions); Deco et al. (2024, psilocybin and escitalopram rebalance brain dynamics via different mechanisms); Siegel et al. (2024, longitudinal precision fMRI showing persistent reduction in anterior-hippocampus to DMN connectivity). Precursor papers (Carhart-Harris 2014 entropic brain; Tagliazucchi 2014 wider repertoire; Lord 2019 metastable exploration; Luppi 2021 LSD integration-segregation) establish the landscape-diversity claim but do not themselves establish explicit restoration.

A critical three-level dissociation caveat attends this subsection. Ort et al. (2023) showed that psilocybin increases spontaneous state-sequence diversity without increasing perturbational complexity (PCI); Casali et al. (2013) and Sarasso et al. (2015) established that spontaneous LZc and PCI measure different things at the consciousness-research level (spontaneous LZc asks "how diverse is the trajectory at rest?", PCI asks "how rich is the response when forced to transition?"). These are not the same observable. Paper 14 specifies that residence-pathology framing targets the dwell-time observable, which is most closely related to Casali-PCI "transition capacity" than to spontaneous LZc "rest diversity." Psychedelics acutely change spontaneous state diversity (Level 1), probably change plasticity substrate (Level 3, Nardou), but do not necessarily change perturbational complexity in the same way (Level 2). The framework claims residence-pathology at the dwell-time-and- transition-capacity level, and does not conflate this with rest- trajectory entropy.

8.6 First-person and public-epistemic substrates

Two first-person-scale instances of the residence-pathology principle warrant treatment in this paper. A third instance (intersubjective coupling on the fastest conversational timescale) is reserved for a follow-on paper on embodied-PCI dynamics where empirical maturity (dual-EEG inter-brain phase-coupling literature) can support a primary-data presentation. Paper 12 v1.4 established the substrate-level demonstration of the residence-pathology pattern at the phenomenological level (the §3.5 smooth-interior / §3.6 boundary-KKT topology, with flow-states characterized by brief boundary touches and intact return to interior, contrasted with

rumination and depression characterized by boundary-residence without recovery). The two subsections below add the memory-substrate instance, which is load-bearing for the framework, and the public-epistemic substrate, which extends the principle to information ecosystems.

8.6.1 Memory as dimensional tether. Memory retrieval is not playback from storage; it is *product-space coupling* between present coherence and past echo. The reconsolidation literature (Nader, Schafe & LeDoux 2000, *Nature* 406:722; Schiller et al. 2010; Agren et al. 2012) established experimentally that retrieved memories enter a labile state and must be re-stabilized by new protein synthesis, modified by the current state during retrieval. The constructive-memory tradition (Schacter, Addis & Buckner 2007; Barsalou 2008; earlier Bartlett 1932) consolidates the evidence that memory retrieval is simulation rather than playback. The false-memory paradigm (Loftus & Palmer 1974) is the structural proof: false memories are insertable by influencing retrieval conditions, which would be impossible if retrieval were factor-space playback. Predictive-processing accounts (Henson & Gagnepain 2010; Friston extensions) and Damasio's (1999) autobiographical-self architecture complete the picture: the self that persists through time is a continuously renewed coupling between present coherence and residual echo structure, not a stored entity.

This treatment connects to orthodox foundations-of-physics work on time as a coupling-derived rather than backdrop phenomenon (Page & Wootters 1983, *Phys Rev D* 27:2885; recent extensions to gravitational time dilation and 3+1 spacetime emergence 2022–2025). The claim that time itself is relational appears in physics foundations and appears in the phenomenology of memory; the framework treats these as coordinate instances of the coupling-through-time principle.

Four memory-coupling pathology subtypes fit within the framework: grief (forced un-coupling where the coupling target has lost its other pole), nostalgia (over-coupling to a past axis with canalized retrieval valence), regret (anti-coupled coupling held against current-axis), and post-traumatic stress disorder (canalized retrieval coupling where each re-triggering instantiates the product-space of present-threat-detection and past-echo, deepening the canal with each retrieval). Each maps to the two-axis observable: $\tau_R(s)$ is dwell-time in the retrieval-coupled state s ; π_c is the occupancy of the retrieval compartment c (present-axis grounded retrieval versus past-axis dominated retrieval). The reconsolidation-window therapeutic literature (Schiller et al. 2010 on fear-memory updating; Ecker et al. 2012 on coherence-therapy reconsolidation) operationalizes intervention as opening a brief window during which the coupling can be re-formed against an updated present coherence — strategy E of §5 (restoring return-path machinery) at the memory substrate.

8.6.2 Public-epistemic substrate — information-ecosystem attractor dynamics. The framework's principle extends to information-ecosystem substrates where the attractor-landscape geometry is shaped collectively rather than individually. Three well-studied empirical literatures provide the substrate's primary anchors. Vosoughi, Roy & Aral (2018, *Science* 359:1146–1151) demonstrated that false news spreads farther, faster, deeper, and more broadly than true news across all categories on Twitter, with the asymmetry strongest for political news; rumour cascades reach deeper retransmission generations and broader unique-user reach than verified content. Cinelli et al. (2020, 2021 PNAS) characterized the “echo-chamber effect” on social media as residence in cluster-segregated interaction patterns where

user opinions and information sources reinforce one another, with platform-architecture differences modulating the strength of segregation. Bail et al. (2018, *PNAS* 115:9216-9221) showed that exposure to opposing political views on social media can *increase* polarization rather than reduce it, identifying the attractor's self-reinforcing dynamics under interventions that naïvely target the residence pattern.

The framework reads these as instances of public-epistemic residence-pathology under the two-axis observable. States s are interpretive frames; compartments c are information-source clusters. $\tau_R(s)$ is the dwell time on an interpretive frame before update under contradicting evidence; π_c is the occupancy probability of source-clusters segregated from contradicting evidence. Pathology is residence in interpretive frames that fail to update under base-rate-relevant evidence (Vosoughi et al. measure this as the cascade-depth advantage of false content), and residence in source-compartments insulated from update (Cinelli et al. measure this as echo-chamber segregation indices). Return-path machinery at this substrate is the set of institutional and cognitive structures that route attention from one source-cluster to another and that re-open closed interpretive frames against contradicting evidence (cross-cutting media consumption, deliberate exposure to source-diverse evidence, institutional fact-checking). When the return-path machinery fails — as the Bail et al. result demonstrates can happen even under nominal cross-exposure — the residence pattern is preserved or deepened.

Financial-market panic dynamics provide a parallel empirical literature for the same residence-pathology pattern at a related substrate. The econophysics literature on volatility clustering and bubble-and-crash dynamics (Mantegna & Stanley 1999; Sornette 2003) characterizes residence-pathology in market-attention attractors with two decades of quantitative methodology that the two-axis observable inherits. We cite these as further empirical instances of the public-epistemic substrate's broader pattern.

8.7 AI / data substrate — model collapse as return-path failure

Shumailov et al. (2024, *Nature* 631:755-759, doi:10.1038/s41586-024-07566-y) demonstrated empirically that generative models trained recursively on data produced by previous model generations undergo *model collapse*: tails of the original distribution disappear, learned behaviors converge to a point-estimate with very small variance, and the process is universal across GMMs, VAEs, and LLMs. Theoretical follow-up (Ren 2024, arXiv:2410.12954) established that this is a statistical phenomenon that may be unavoidable under recursive generative training in the absence of corrective real-data circulation.

Wu, Kang, Xu, Xie, Mi, Wang, Liu & Chen (2026, arXiv:2605.06347, “Human-AI Co-Evolution and Epistemic Collapse: A Dynamical Systems Perspective,” posted 7 May 2026) extended this analysis from model-internal recursive training to *coupled* human-AI dynamics, modeling humans and language models as a single coupled dynamical system with three variables (human cognitive capacity H , data quality Q , model capability M) and feedback loop $H \rightarrow Q \rightarrow M \rightarrow H$. They identify three regimes: co-evolutionary enhancement, fragile equilibrium, and degenerative convergence, with the system transitioning toward a low-diversity suboptimal equilibrium as cognitive-offloading increases. From an information-theoretic perspective, the degenerative regime corresponds to an emergent information bottleneck along the feedback loop, with entropy reduction reflecting loss of diversity rather than beneficial compression. This proposal — that human-AI coupled dynamics exhibit three qualitatively distinct

attractors with the degenerative one driven by recursive feedback failure — is the minimal-model version of the framework’s residence-pathology principle at the AI/data substrate.

The framework reads both findings as instances of return-path-machinery failure on the AI/data coupling substrate. The “return path” at this substrate is circulation of non-synthetic, world-grounded data and intact human cognitive engagement into the training distribution. When recursive synthetic generation dominates and cognitive offloading rises, the return path fails; the coupled system residences in a degraded attractor with reduced variance and lost diversity. This is structurally identical to canalization at the psychiatric substrate (chronic residence in deepened attractor canals), aggregation-pathology at the molecular substrate (chronic residence in pathological conformational states), and operator-sustained boundary operation at the cardiac substrate (chronic residence past return-machinery cutoff). The same $R(s,c)$ two-axis observable applies: states are model-distribution configurations; compartments are training-data regimes; pathology is residence in degraded configurations with reduced support and lost tail mass.

The framework’s therapeutic prescription at this substrate — restore the return-path circulation, audit the coupling, refuse the singularity attractor — is consistent with the mitigation strategies proposed independently by Wu et al.: improved data curation, human-in-the-loop system design, and educational practices that preserve active cognitive engagement. The framework adds: these are not separate interventions but coordinate instances of return-path-machinery restoration, with a common observable (recovery time constant of the coupled system’s response to perturbation).

8.8 Synthesis across the extensions

Figure 2 presents the cross-substrate grammar in a single matrix: eight substrates, six columns (state, compartment, τ_R observable, π_c observable, return-path machinery, canonical pathology), one principle.

Eight substrate instances — pharmacology, cancer phospho-regulation, autonomic physiology, exercise physiology, brain-network dynamics, memory phenomenology, public-epistemic information ecosystems, AI/data — one principle, one observable, one symptom grammar. At each substrate the residence-pathology structure applies: a system with two regimes, explicit return-path machinery, and measurable recovery kinetics can fail when pathological residence accumulates past design-intent bounds. The two-axis observable $R(s, c) = \tau_R(s) \times \pi_c$ captures the measurable signal; the return-path-machinery typology (§6) captures the mechanism of restoration; the frustration-without-escape extension (§7) captures the most precise formal statement at the proteostatic substrate. The cross-substrate claim is not metaphor. Each substrate has its own independently developed vocabulary for residence patterns; the framework names the uniformity that those vocabularies collectively describe.

9. Differentiation from Adjacent Literature

This section states, for each adjacent body of work, what it claims, what we concede to it, and what remains specific to the present paper. The goal is to preempt the most plausible reading that this paper is restating something already published.

9.1 Protein-biophysics lineage

Knowles, Vendruscolo & Dobson (2014, 2017) — protein metastasis. The phrase “protein metastasis” already names spreading of conformational pathology across cellular compartments and across cells. We concede the metaphor and the proteostatic instance. The Knowles-Vendruscolo-Dobson program does not generalize across substrates and does not propose a two-axis residence-time observable; it operates within the protein-aggregation field. Our contribution is the substrate-general formal observable and the symmetric formulation of kinetic stabilization (§5).

Kelly (1996) and the kinetic-stabilization literature. Kelly’s foundational work on transthyretin stabilization established that extending the dwell time of the native fold (strategy A in §5) is a tractable therapeutic strategy. Tafamidis is the canonical clinical realization. We retain Kelly’s framing as one of five symmetric residence-control strategies and explicitly cite it as prior art for strategy A. Strategies B through E (destabilize pathological dwell; reroute away from pathological compartment; reroute toward protective compartment; restore return-path machinery) are formulated symmetrically with strategy A; the symmetric formulation as a complete strategy set is the framework’s incremental contribution at the molecular substrate.

Pytel, Fromm & Longo (2025) and the Wolynes-Ferreiro frustration program (Parra, Komives, Wolynes & Ferreiro 2025 review). Frustration theory characterizes which interactions in a folded protein are energetically frustrated. The Pytel-Fromm-Longo 2025 work and the Parra-Komives-Wolynes-Ferreiro review extend the frustration vocabulary to disordered regions and condensate biology. Frustration as published is state-shaped (which interactions are frustrated); the framework’s extension to *frustration without escape* is residence-shaped (how long the frustrated state persists). We claim this extension as a specific contribution and propose courtesy preprint to Ferreiro and Wolynes prior to journal submission per the framework’s preprint-circulation strategy.

Jülicher, Weber and the active-matter / biomolecular-condensate program. Active-matter theory and the biomolecular-condensate field (Hyman, Brangwynne, and collaborators) provide the physical-chemistry vocabulary for liquid-liquid phase separation, condensate aging, and gel transition. We treat this as prior art for the FUS-stress-granule case (§3.1) and adopt its vocabulary. The framework does not contribute new condensate physics; it contributes the cross-class residence-pathology grammar that subsumes condensate pathology as one instance.

9.2 Psychiatric / brain-network lineage

Carhart-Harris REBUS and canalization (Carhart-Harris & Friston 2019; Carhart-Harris, Chandaria, Erritzoe, Gazzaley, Girn, Kringelbach, Kuypers, Leech, McCulloch, Moliner, Olsen, Robbins, Robertson, Rolls, Sahakian, Schartner, Searle, Singh, Stamets, Timmermann, Tagliazucchi, Trickett, Tyts, Watts, Whalley, Williams & Nutt 2023). The canalization-theory program established that residence-in-deepened-attractor-canals organizes a wide range of psychiatric disorders, and that psychedelic interventions act by transiently flattening the energy landscape to enable transitions out of pathological canals. This is the closest published prior art to our cross-substrate claim, and it closes the brain-level case completely. We do not claim cross-disorder residence-pathology at the brain level. We claim

formal-operational composition of the brain-level case with substrate-distinct cases (molecular, autonomic, cardiac, pharmacological, AI/data) under a single observable.

Default-mode-network hyperstability literature (Hamilton et al. 2011; Sheline et al. 2009, 2010; Wise et al. 2017; Kaiser et al. 2016; Alonso et al. 2022). The empirical DMN literature supports maladaptive persistence and impaired switching under demand, not naive static hyperconnectivity. We adopt the careful framing (§8.5): residence in DMN-dominant states with impaired return-to-task-set, not static hyperstability.

9.3 Pharmacology lineage

Copeland residence-time paradigm (Tummino & Copeland 2008; Copeland 2016). The drug-target residence-time literature is the most mature published instance of residence-pathology thinking. We adopt its vocabulary explicitly, cite the Bosma et al. 2017 H1 antihistamine data and Casarosa et al. 2009 M3 LAMA data as the empirically strongest cases for residence-time dominance over equilibrium affinity, and propose that the four-point audit-design template (§10) is a substrate-general adaptation of Copeland's pharmacological audit-design. The pharmacological case is conceded as the canonical published precedent; our contribution is generalization beyond drug-target to substrate-neutral coupling.

9.4 Process-relational philosophy and relational ontology

Rovelli relational quantum mechanics (RQM; Rovelli 1996, 2018, 2021). RQM proposes that quantum-mechanical properties are inherently relational, defined only with respect to an observer-system pair. At the conceptual level, this is the closest physics-foundations prior art for a coupling-first observable in a fundamental-physics substrate. We concede the relational ontology at the quantum-foundations level. The framework's two-axis residence-time observable $R(s,c)$ does not require RQM; the framework's claim is empirical and operational rather than interpretational.

French and Ladyman ontic structural realism (OSR; Ladyman 1998; French 2014). OSR holds that the fundamental ontology of physics is relations rather than relata. We concede OSR as the closest published prior art for substrate-neutral relational ontology at the level of philosophy of physics. The framework does not depend on OSR but is structurally compatible with it.

Barad agential realism (Barad 2007). Agential realism specifies that observables are constituted by apparatus-specific intra-action between phenomenon and measurement-arrangement. This is *apparatus-specific* observable machinery; our framework is *substrate-general* observable machinery. The two programs are compatible at the level of methodology — both reject substance-first ontology — but they answer different questions. Agential realism specifies how an observable comes into being; the present framework specifies what observable is operationally measurable across substrates.

Whitehead process philosophy (Whitehead 1929; Stengers 2011). Whitehead's process metaphysics is the canonical philosophical precedent for relation-and-occasion as the fundamental ontological category. Lane CP-1 returned a modular-preemption verdict: process philosophy preempts the *conceptual* foundation of coupling-first ontology, but the formal-operational layer (a measurable cross-substrate observable with substrate-distinct

realizations) remains open. We retain Whitehead's actual-occasions-as-units as a philosophical influence and cite explicitly; we do not claim first articulation of process-relational ontology.

Simondon individuation theory (Simondon 1958/2020). Simondon's program treats individuation as the primary ontological act and substantive individuals as derivative. The formalization of Simondon's framework is incomplete in the published literature; we cite it as a philosophical companion and do not depend on its formal completion.

9.5 Biological / cognitive-science lineage

Maturana and Varela autopoiesis (Maturana & Varela 1980; Varela, Thompson & Rosch 1991; Thompson 2007 *Mind in Life*). Autopoiesis specifies that the living is constituted by self-producing relational networks of components. Thompson's *Mind in Life* extends the program to phenomenology and consciousness. Lane CP-3 returned: this lineage closes the *biological* substrate's coupling-first ontology completely. We do not claim novelty in biological coupling-first ontology. Our contribution is the formal-operational integration across biological and non-biological substrates.

Friston free-energy principle (FEP; Friston 2010, 2019). FEP is a methodological framework for inferring belief-update dynamics from observable behavior, with active-inference extending it to action selection. Lane CP-3 verdict: FEP stops at methodological / inferential. It does not assert that the substrate of living systems is coupling rather than substance; it assumes a system-environment partition and computes belief dynamics over it. The framework is compatible with FEP but does not depend on it; we do not claim FEP-style derivation of the residence-pathology principle from variational arguments.

Rosen relational biology (Rosen 1991, 2000). Rosen's framework distinguishes organisms as relational (M,R)-systems rather than mechanistic. We concede the conceptual primacy of relational biology for the biological substrate and treat it as part of the closed biological case.

Damasio (1999) autobiographical-self architecture; Tononi-Edelman dynamic-core / IIT. These provide vocabulary for self and consciousness at the cognitive substrate. The framework is compatible with both and does not claim to displace either. The R(s,c) observable could in principle be measured against IIT-style integration indices in future empirical work.

9.6 Engineering and dynamical-systems lineage

Engineering resilience and critical slowing down (van Nes & Scheffer 2007; Scheffer et al. 2009; van de Leemput et al. 2014). The dynamical-systems literature on resilience and critical slowing down provides the mathematical vocabulary for recovery time constants near bifurcations. We adopt this vocabulary explicitly for the common observable of return-path-machinery (§6.2) — recovery time constant τ , dominant eigenvalue, dwell-time persistence. The framework's contribution is naming this vocabulary as the substrate-general language for return-path quality, not the vocabulary itself.

Lipsitz and Goldberger loss-of-complexity hypothesis (1992). Loss-of-complexity proposes that aging and disease entail loss of physiological complexity. This is conceptually adjacent to return-path-machinery failure; we cite as compatible prior art. The two-axis residence-time

observable refines the loss-of-complexity claim by distinguishing residence-axis pathology (dwell-time persistence) from transition-axis pathology (variety / entropy of transitions).

Page-Wootters mechanism (Page & Wootters 1983; recent 2024-2025 extensions to gravitational time dilation). The Page-Wootters mechanism derives time-as-coupling between a clock subsystem and a system-of-interest within a global timeless quantum state. We cite Page-Wootters as orthodox physics-foundations prior art for the claim that time is coupling-derived rather than backdrop, supporting the memory-as-dimensional-tether subsection (§8.6.1). This is referenced for theoretical context only; the empirical observable $R(s,c)$ does not require Page-Wootters.

9.7 The specific contribution restated

Given the above concessions: this paper's contribution is not the relational ontology at any single substrate, not the residence-time concept at any single substrate, not cross-disorder residence-pathology at the brain level, and not the recovery-time-constant vocabulary. The contribution is the substrate-general two-axis observable $R(s, c) = \tau_R(s) \times \pi_c$ together with the symmetric kinetic-stabilization formulation (§5), the return-path-machinery typology (§6), the frustration-without-escape extension (§7), and the demonstration that one observable plus one symptom-grammar resolves residence-pathology across at least eight independently developed substrate vocabularies (§8). This is what is meant by *formal-operational composition*: not new substrate physics at any one level, but the explicit measurable composition rule across levels that prior programs developed in isolation.

10. Audit-Design Template

This section adapts the Copeland four-point pharmacological audit-design (Tummino & Copeland 2008; Copeland 2016) to a substrate-general residence-pathology audit-design. The intent is that any empirical claim about residence-pathology at any substrate should be required to report, in publication, on each of the four points below at the appropriate substrate-specific level.

10.1 The four required reports

(1) Full two-axis occupancy quartet. For each system under study, report (a) state populations — the probability mass at each operationally distinguished state s ; (b) dwell times — the residence time $\tau_R(s)$ at each state, ideally as a distribution rather than a point estimate; (c) compartmental occupancy probabilities — π_c for each compartment c ; (d) transition rates — the rate matrix governing transitions between (s, c) pairs. At the molecular substrate this is the kinetic quartet (state populations, conformational dwell times, compartmental fractions, on/off rate constants). At the brain-network substrate this is the dynamics-of-state-sequences quartet (state populations of metastable networks, dwell times in each metastable network, region-occupancy probabilities, transition probabilities between states). At the AI/data substrate this is the distribution quartet (output-distribution mass at each mode, dwell times in each mode under iterated inference, compartmental segregation of training-data populations, transition rates between modes under prompting perturbation).

(2) Washout-sensitive functional assay. Pair the occupancy quartet with a functional readout that responds to withdrawal of the active perturbation. At the pharmacological

substrate this is the receptor-recovery time constant after drug washout. At the cardiac substrate this is HRV recovery after stressor withdrawal. At the proteostatic substrate this is the persistence of stress-granule disassembly after stressor removal. At the brain-network substrate this is the return-to-task-set time constant after cognitive load. The principle: residence-pathology is not detectable by equilibrium readout alone; it requires a withdrawal-recovery measurement.

(3) In vivo / in-tissue persistence after acute trigger removal. Demonstrate that the residence pattern persists in the native context after the proximate trigger is removed. This is what distinguishes pathological residence (system remains in the abnormal compartment after the stressor / agent / context that produced it is gone) from healthy transient response (system returns to baseline as soon as the trigger is removed). The temporal scale of “persists” depends on the substrate but must be specified.

(4) Local temporal and spatial modifiers. Report on the local context that modifies residence behavior: concentration, post-translational modifications, cofactor availability, membrane environment, micro-compartmental context, network-state, hormonal state, autonomic baseline, data-distribution conditioning. Residence-pathology is substrate-specific and context-sensitive; an audit that reports residence times without reporting modifiers cannot be cross-substrate compared.

10.2 Substrate-general version

For any new substrate proposed to exhibit residence-pathology under the framework’s grammar, the audit-design template requires: full kinetic quartet at the appropriate level (states, dwells, compartmental occupancies, transition rates) + washout-sensitive assay (functional readout after withdrawal of active perturbation) + in situ persistence (native-context residence after proximate-trigger removal) + local micro-context (modifiers of residence behavior). When all four are reported, the system is admissible for cross-substrate comparison under $R(s, c)$. When any are missing, the claim of residence-pathology at that substrate is provisional.

10.3 Standards adoption

We propose this template as a publication standard for residence-pathology claims, parallel to how pharmacological residence-time work has matured toward standardized reporting (Copeland audit-design, Bosma et al. 2017 audit-design as exemplary). The standardization is what makes cross-substrate composition empirically tractable rather than metaphorical.

11. Cross-Domain Experimental Agenda

Figure 5 organizes the thirteen experiments below by cluster, tractability tier, and dependency.

Thirteen experiments are proposed below, organized into two clusters: cross-substrate recovery-kinetics experiments (eight; from Lane 1) and pharmacology-to-psychedelics bridge experiments (five; from Lane 4). Each experiment is designed to falsify a specific component of the framework. All are tractable with current technology except where noted.

11.1 Cross-substrate recovery-kinetics cluster

Experiment 1: Cross-domain perturbation-recovery battery. Recruit a single cohort and measure recovery time constants τ across at least four substrates per subject: (a) cardiac — HRV recovery after standardized cold-pressor or orthostatic stressor; (b) proteostatic — peripheral-blood stress-granule disassembly kinetics after sodium-arsenite challenge in isolated PBMCs; (c) brain-network — fMRI return-to-task-set after working-memory load; (d) phospho-regulatory — GSK3 β /PP2A phospho-substrate settling time after lithium washout in lymphoblast lines. The prediction: within-subject τ across substrates will be positively correlated (subjects with slower return on one substrate will have slower return on others) at $r > 0.4$. Falsification: substrate-specific τ are independent ($r < 0.2$ across all pairs). Power calculation suggests $n = 80$ for adequate cross-correlation estimation.

Experiment 2: Multimodal HRV + fMRI in depression. In a depressed cohort ($n = 60$) versus controls ($n = 60$), measure (a) resting HRV plus HRV recovery from cold-pressor; (b) fMRI dwell-time distributions for default-mode-network metastable states plus return-to-task-set time after working-memory load. Prediction: depression severity correlates with HRV recovery τ and with DMN return-to-task-set τ *separately and additively*, and these two measures jointly predict treatment response better than either alone. Falsification: only one or neither correlates with severity / response.

Experiment 3: Proteostasis-phosphoregulation coupling assays. In a tractable cell-line model (HEK293 or iPSC-derived neurons), induce proteostatic stress (sodium arsenite) and measure phospho-substrate kinetics for known PP2A and GSK3 β substrates in parallel with stress-granule formation / disassembly. Prediction: phospho-regulatory return-path failure (slowed phospho-substrate resetting after stressor washout) co-occurs with proteostatic return-path failure (slowed granule disassembly), with shared dominant eigenvalue across the two substrates. Falsification: the two return-path failures are statistically independent.

Experiment 4: Dominant-eigenvalue modeling of recovery. For each substrate in Experiment 1, fit a low-dimensional dynamical-systems model (linearized around baseline) to perturbation-recovery time-courses; extract the dominant eigenvalue. Prediction: within-subject dominant eigenvalues are correlated across substrates at the same level as the bulk recovery time constants in Experiment 1, supporting the claim that recovery quality is governed by a substrate-specific instance of a substrate-general principle. Falsification: dominant eigenvalues are statistically independent of bulk τ within substrate.

Experiment 5: Longitudinal early-warning designs. Recruit a high-risk cohort for a defined transition (e.g., subjects with first-episode major depression for relapse prediction, $n = 200$, two-year follow-up). At baseline and quarterly, measure substrate-specific recovery τ across at least three substrates. Prediction: rising τ across substrates over time predicts the transition (relapse) with longer lead-time than any single-substrate measure. Falsification: cross-substrate composite does not improve over single-substrate prediction.

Experiment 6: Return-path composite index. Develop and validate a composite index combining standardized HRV recovery, fMRI dwell-time persistence, and a peripheral-blood proteostatic recovery readout. Prediction: the composite outperforms each component in differentiating clinical groups (depression, anxiety, post-stroke depression, frontotemporal

dementia) from controls, with effect-size improvement ≥ 0.3 in Cohen's d over the best single component. Falsification: composite does not outperform best single component.

Experiment 7: Intervention-convergence analysis. Subjects with confirmed slow recovery on the composite index (Experiment 6) are randomized to four interventions: (a) HRV biofeedback / tVNS; (b) cognitive training targeting return-to-task-set; (c) low-dose lithium (phospho-regulatory return-path); (d) waitlist control. Prediction: all three active interventions improve the composite index at six-week post-randomization, with the substrate-specific intervention improving its target substrate most but with measurable cross-substrate transfer in all three. Falsification: each intervention is purely substrate-specific with no cross-substrate transfer.

Experiment 8: Boundary-condition experiments. Identify clinical populations where return-path machinery is intact but residence-pathology occurs through an alternative mechanism (e.g., genetic FUS mutations producing granule pathology without altered recovery machinery; pure-OCD with intact DMN switching but residence-dominated symptom phenomenology). Prediction: $R(s, c)$ measurements in these populations show pathological residence on one axis with intact recovery time constants on the other, falsifying any simple "return-path-failure-only" reading of the framework. This is a self-falsification experiment: it tests whether the two-axis observable correctly identifies residence-pathology that does not reduce to return-path failure.

11.2 Pharmacology-to-psychedelics bridge cluster

Experiment 9: Matched-exposure varied- k_{off} 5-HT_{2A} agonist panel. Develop or identify a panel of 5-HT_{2A} agonists with matched receptor occupancy time-courses but varied k_{off} (residence time at the receptor). Prediction: subjective and neuroimaging readouts of the canalization-flattening effect scale with k_{off} -derived residence time rather than peak occupancy. Falsification: equal-occupancy agonists with different k_{off} produce equivalent canalization-flattening readouts.

Experiment 10: Antagonist-termination experiment (killer experiment). This experiment requires no new technology and directly tests the residence-pathology reading of psychedelic action. Administer psilocybin under standard dosing; at a pre-specified time-point during the experience (e.g., $t = 90$ min), randomize subjects to placebo or ketanserin (5-HT_{2A} antagonist with fast on-rate). Prediction: ketanserin truncates the receptor-residence pattern and produces measurable shortening of the post-acute neural-dynamic effects in proportion to the duration of residence interruption. Falsification: ketanserin terminates the acute experience but post-acute neural-dynamic effects are unaffected, indicating that residence-on-target is not the relevant variable.

Experiment 11: Clinical occupancy-to-dynamics PK/PD model in psilocybin depression trial. Within an active psilocybin-for-depression trial, develop a PK/PD model linking serum psilocin time-course to receptor occupancy time-course to neural-dynamic readouts (HMM dwell times, return-to-task-set τ) to clinical response. Prediction: the residence-time-derived predictor (integral of occupancy over time) outperforms the peak-occupancy predictor and the dose-only predictor in explaining clinical response variance. Falsification: residence-time predictor does not add explanatory value over peak occupancy.

Experiment 12: Spontaneous-versus-perturbational dissociation study. Simultaneously measure spontaneous-state-sequence complexity (HMM-derived metrics from resting-state fMRI), perturbational complexity (PCI from TMS-EEG; Casali et al. 2013; Sarasso et al. 2015), and long-term plastic-reopening markers (Ort 2023 paradigm) in a psychedelic-intervention cohort. Prediction: these three readouts dissociate — they correlate within each subject at $r < 0.5$ across the three pairs — supporting the framework’s claim that residence-pathology and return-path-machinery operate on distinct measurable channels rather than reducing to a single complexity readout. Falsification: the three readouts correlate at $r > 0.8$ (collapsing to one measure).

Experiment 13: Baseline-rigidity-as-moderator design. Pre-stratify a psychedelic-intervention cohort by baseline rigidity of state-sequence dynamics (HMM dwell-time persistence, fMRI return-to-task-set, HRV recovery τ). Prediction: subjects with greater baseline rigidity show larger absolute improvements in both the residence readout and clinical response, supporting the framework’s prediction that the residence-pathology dimension explains heterogeneity of response. Falsification: baseline rigidity does not moderate response.

11.3 Tractability summary

Experiments 1, 3, 4, 8, 10, 12, 13 are tractable with current technology in mid-sized academic-cohort designs ($n = 60$ -200). Experiments 2, 5, 6, 7, 11 require larger cohorts or multi-site coordination but do not require new instrumentation. Experiment 9 requires medicinal-chemistry pre-work to develop or identify the matched-exposure varied- k_{off} panel and is therefore the most tractability-bounded of the agenda.

11.4 Sample-size justifications

Sample-size targets specified above derive from standard power-analytic considerations applied to the framework’s primary effect-size predictions. The justifications below are summary; full power-analysis worksheets accompanying any pre-registration of each experiment will be deposited alongside the registration.

Experiment 1 ($n = 80$). The cross-substrate correlation prediction is $r > 0.4$ (Pearson) on each pairwise comparison among at least four substrate-specific recovery time constants. With $n = 80$, a single pairwise correlation has 80% power to detect $r = 0.4$ at $\alpha = 0.05$ two-sided. The four-substrate design produces six pairwise correlations; under the multiple-comparison correction appropriate for this exploratory design (Benjamini-Hochberg, $q = 0.05$), $n = 80$ remains adequate for the *pattern* prediction that the majority of pairwise correlations exceed $r = 0.4$ with the average across pairs significantly different from zero. The cross-correlation matrix structure is the primary test, not any individual pair.

Experiment 2 ($n = 60 + 60$). The depression-versus-control comparison on HRV recovery τ typically shows Cohen’s d in the 0.5–0.7 range (per published HRV-and-depression meta-analyses); on DMN return-to-task-set τ typical effect sizes are $d = 0.4$ –0.6. With $n = 60$ per group, two-sided independent-samples t-tests have 80% power to detect $d \geq 0.52$ at $\alpha = 0.05$. The additivity-of-predictors hypothesis is the primary test (incremental R^2 from adding the second predictor) and requires the same total sample for adequate power under typical multimodal-prediction effect sizes.

Experiment 5 (n = 200, two-year longitudinal). For relapse-prediction at 24 months in first-episode major depression with ~40% expected relapse rate (concordant with FE-MDD longitudinal literature), n = 200 yields ~80 events. Cox regression with a single composite-index predictor (cross-substrate τ -composite) requires roughly 10–20 events per predictor for stable estimation; n = 200 supports a composite predictor plus 3–4 covariate adjustments. Comparison of cross-substrate composite to single-substrate predictors uses nested-Cox c-index improvement with non-parametric bootstrap, for which n = 200 has been shown sufficient in cardiovascular-risk-composite literature.

Experiment 6 (n = 50 per clinical group, four groups + controls). The Cohen's $d \geq 0.3$ effect-size improvement target requires $n \approx 50$ per group for 80% power at $\alpha = 0.05$ two-sided when comparing the composite-index group-separation against the best single-substrate component using paired comparisons within subjects. The four-clinical-group design (depression, anxiety, post-stroke depression, frontotemporal dementia) supports both omnibus and pairwise differentiation tests.

Experiment 7 (n = 30–40 per arm, four arms). The intervention-convergence prediction is that all three active arms improve the cross-substrate composite at six weeks post-randomization, with effect-size pattern of substrate-targeted > cross-substrate-transfer > none in waitlist. With n = 35 per arm (total n = 140), 80% power exists to detect $d = 0.6$ within-arm pre-post improvement at $\alpha = 0.05$; cross-arm pattern tests via mixed-effects regression are powered for the predicted ordinal pattern under realistic effect-size assumptions.

Experiment 11 (PK/PD model within active trial). Sample size is bounded by the host trial; the PK/PD secondary-analysis design typically requires $n \geq 30$ with serial sampling for stable parameter estimation, achievable within current psilocybin-for-depression trials. The residence-time-derived predictor's incremental explanatory value over peak-occupancy predictor is the primary test, evaluated by likelihood-ratio in the nested PK/PD model.

Experiments 3, 4, 8, 9, 10, 12, 13 have their sample-size targets governed by substrate-specific or assay-specific considerations rather than by cohort-level effect-size detection; the host design (cell-line replicates, animal-model n, single-arm psychedelic-cohort n) determines the achievable power, and pre-registration worksheets will specify these per experiment at the time of registration.

11.5 Pre-registration commitment

Experiments 2, 5, 6, 7, and 11 will be pre-registered on the Open Science Framework (OSF) prior to data collection. Pre-registration will specify primary and secondary endpoints, planned analyses, sample-size justifications including the worksheets summarized in §11.4, planned subgroup analyses, and the falsification conditions stated in each experiment's main-text description. Experiments 1, 3, 4, 8, 10, 12, and 13 will be pre-registered when their substrate-specific designs are finalized in collaboration with appropriate domain laboratories; for the cell-line and TMS-EEG experiments, this means at the point of laboratory-specific protocol agreement rather than at the design-level proposal stage of this paper. Experiment 9 cannot be pre-registered at the cross-design level until the matched-exposure varied- k_{off} agonist panel exists; medicinal-chemistry pre-work is the rate-limiting step.

The pre-registration commitment is part of the framework's broader audit-design discipline (§10): residence-pathology claims should be evaluated against pre-specified falsification conditions rather than against post-hoc reframing. The experimental agenda is offered as a falsification program, not as confirmation theater.

12. Implications

12.1 Therapeutics

The framework reframes therapeutic intent across substrates as residence-control rather than as substance-control. Restore return-path machinery wherever it has failed; abolish pathological residence on either the state-axis or the compartment-axis. The five symmetric strategies (§5) — extending native dwell, destabilizing pathological dwell, rerouting away from pathological compartment, rerouting toward protective compartment, restoring return-path machinery — organize the design space for any specific intervention.

The most important translational caution must be stated explicitly: identifying the principle correctly does not guarantee that any specific intervention based on it will work. Target engagement, timing, leverage, and substrate-specific physiology all matter independently. The arimoclomol ORARIALS-01 phase 3 trial in ALS is a warning case: a residence-targeting intervention with strong mechanistic rationale failed at phase 3 despite passing earlier trials, and the failure cannot be read as falsifying the residence-pathology framework, only as demonstrating that mechanistic correctness is necessary-but-not-sufficient for clinical efficacy. This warning generalizes — the framework predicts where to look, not what to do.

Intervention-convergence (Experiment 7) provides the empirical handle: if substrate-specific interventions converge on the cross-substrate composite readout, the residence-pathology principle is corroborated even when any single intervention fails. The principle does not stand or fall on the success of any one drug, biofeedback protocol, or stimulation paradigm; it stands or falls on whether residence-pathology measurements track clinical course and intervention response in the predicted way.

12.2 Diagnostics

Measure $R(s, c)$ and recovery time constants before structural aggregation markers, which are downstream. The framework predicts that residence-pathology readouts will outperform structural markers at early and pre-symptomatic stages of disease. This has direct application to neurodegenerative disease, where structural markers (plaques, tangles, aggregates) are by the time of detection already late-stage manifestations of years-to-decades of upstream residence-pathology. The same logic applies to depression (early-warning before relapse; Experiment 5), to autonomic / cardiac disease (HRV degradation as years-earlier warning of cardiovascular event; conceded from Framingham), and to cognitive aging (return-path-machinery composite as years-earlier warning of clinical decline).

The Substrate-General Audit-Design Template (§10) gives the publication standard for residence-pathology diagnostic claims, intended to mature this clinical use-case in the same way Copeland's pharmacological audit-design matured pharmacology.

12.3 Disease classification

Group disorders by their residence-axis signature plus return-path-machinery failure profile rather than by aggregate morphology or single-time-point biomarker. Concretely: a single subject with depression, family history of cardiovascular disease, and elevated proteostatic stress markers, all converging on slow recovery time constants across substrates, is more usefully classified as a pan-substrate-return-path-failure case than as comorbid-depression-plus-cardiovascular-risk-plus-proteostatic-pathology. The classification has therapeutic implications (multi-substrate intervention with convergent endpoint readouts) and prognostic implications (cross-substrate composite as integrated risk profile).

This is a re-organization rather than a re-naming. The existing diagnostic categories continue to identify subgroups within the cross-substrate composite; the composite organizes the categories. Group-level diagnostic stratification by residence-axis signature provides the missing structural axis for explaining heterogeneity of intervention response and disease trajectory within named categories.

12.4 Research methodology

The framework predicts that cross-substrate experimental designs (Experiments 1, 6, 7) will produce more replicable, more powerful results than single-substrate designs, because cross-substrate composites integrate over substrate-specific noise and reflect the substrate-general residence-pathology dimension directly. If correct, this implies a research-methodology shift: multi-substrate cohorts measured across substrates per subject become the canonical experimental design, with single-substrate work used for mechanism-elucidation rather than for primary endpoints. This methodological shift is the framework's most consequential research-program-level implication.

13. Non-Goals

This paper is not a medical-advice claim. The framework observes structure; clinical work remains the domain of clinical literature. This paper does not claim first articulation of relation-first ontology at any substrate level. It does not claim first articulation of cross-disorder residence-pathology at the brain level (canalization theory established that in 2023). It does not claim the mathematical-physics interpretation of the observable, which is deferred to a companion paper.

14. Acknowledgment of Deferred Companion Paper

The framework's underlying algebraic structure — specifically the interpretation of G_2 symmetry as the symmetry of coherent coupling structures and of octonion non-associativity as expressing irreducible relationality — requires substantial derivation work beyond this paper's scope. We identify that work as belonging in a separate treatment (Graise, in preparation) whose burden is three deliverables: (i) derive why non-associativity should be read as irreducible relationality rather than merely algebraic failure of reassociation; (ii) show why G_2 specifically, rather than some broader exceptional or categorical structure, is the right symmetry notion for coherent coupling; and (iii) define at least one concrete invariant or observable from associators or G_2 -equivariant quantities that distinguishes coherent from

failed coupling. The present paper's observable $R(s, c)$ does not depend on that companion paper's interpretation and is stated here in substrate-neutral terms.

Drafting status and next steps

v1 progress as of 2026-05-12 04:xx PDT:

All fourteen sections drafted in full first-draft prose. Abstract, §1 introduction, §2 formal observable, §3 four canonical disease cases, §4 cross-class unifying claim, §5 symmetric kinetic-stabilization formulation, §6 return-path-machinery typology, §7 frustration-without-escape, §8 cross-substrate extensions (eight substrates plus synthesis), §9 differentiation from adjacent literature, §10 audit-design template, §11 thirteen-experiment cross-domain agenda, §12 implications across therapeutics / diagnostics / classification / methodology, §13 non-goals, §14 deferred-companion-paper acknowledgment.

Remaining work toward v1 final:

- Internal review pass for citation completeness and tightness
- Reference list consolidation — every cited work in the running text needs a complete reference entry at submission time
- Substrate-specific terminology audit (consistency of “residence-pathology”, “return-path machinery”, “coupling-failure”)
- Section-length rebalancing (§8 may be too long; §4 may be too short)
- Figure plan: candidate figures include the two-axis observable schematic, the four-case occupancy quartet visualization, the cross-substrate residence-pathology grammar table, the return-path machinery typology table, and the experimental-agenda dependency graph
- Companion-paper plan: Paper 15 (G_2 / octonion mathematical-physics) outlined separately in outbox/syntheses/lane_cp5_octonion_coupling/ with three deliverable burden specified in §14 above

Review cycle: v1 goes to courtesy-preprint Tier A (11 names) two weeks before submission; Tier B (7 names) one week before; revisions incorporated as v1.1; then journal submission. The differentiation section (§9) is structured to anticipate the most likely critic-readings before they arrive.

Submission targets to evaluate: *Trends in Cognitive Sciences*, *Nature Reviews Drug Discovery*, *Nature Reviews Neuroscience*, *Cell Systems*, *PLOS Computational Biology*. Trade-off between cross-substrate-scope journals (TICS, Cell Systems) and substrate-specific journals (NRDD for the pharmacological / proteostatic emphasis, NRN for the brain-network emphasis); decision to be made after internal-review pass.